Zero-Violation Regret for Cooperative Markov Games with Coupled Instantaneous Hard Constraints

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Abstract

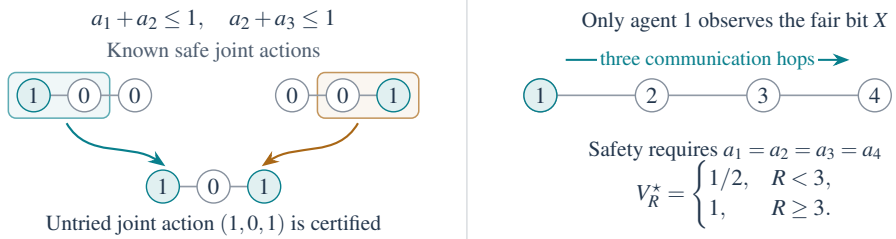
Robotic teams and wireless networks must learn coordinated behavior while respecting collision and interference constraints throughout exploration. Expected cumulative-cost guarantees do not ensure per-step safety, and individually feasible actions can violate coupled constraints when combined. We ask when safe observations and limited communication enable near-optimal cooperation without unsafe exploration. We study episodic cooperative Markov games with unknown linear reward and constraint models, linear-mixture transitions, and coupled instantaneous hard constraints. Agents act using local observations and communication over a prescribed graph. SAFE-CMG-UCBcombines known and learned local safety certificates with optimistic Bellman planning, enforcing consistency across overlapping scopes while retaining a fallback safe at every modeled state. We prove high-probability zero violation and a finite-time regret bound separating model-estimation error from episodes with uncertified optimal actions. Under exact communication-compatible planning and explicit safety-margin and safe-information coverage conditions, the bound yields sublinear regret against the optimal admissible safe protocol. We characterize safety information obtained by local recombination and bound tabular certification complexity by local configuration counts. A constructive distributed protocol quantifies state acquisition and routed communication, while bounded width of the complete planning graph permits efficient per-state action optimization. Impossibility results show that a safe fallback alone need not enable sublinear regret and that insufficient communication can reduce optimal safe value even at treewidth one.

1 Introduction

Robots sharing a workspace and transmitters sharing a channel must coordinate their actions to avoid collisions or excessive interference [1, 2]. Reinforcement learning (RL) can improve their decisions from interaction [3], but collecting data must also be safe. An expected cumulative-cost budget can permit an unsafe step even when the total cost satisfies the budget in expectation. This distinction matters when a single collision or interference event causes harm. We study *instantaneous hard constraints*, which require every executed joint action to satisfy every constraint throughout learning. The objective is high-probability safety and return approaching that of the best admissible safe policy.

Coupled constraints make both learning and execution difficult. Two robot motions that are safe in isolation can collide when executed together. Likewise, several individually permissible transmissions can jointly exceed a receiver's interference limit. Certifying each agent's action independently is therefore insufficient. An agent may also participate in several constraints, and its action must satisfy all of them simultaneously. Which joint actions can be certified from safely collected observations, and how can agents agree on one before execution? Answering these questions requires both informative safety observations and enough communication to coordinate the agents' actions.

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(a) Reusing local safety certificates **(b) Communication limits safe value**

Figure 1: Two structural effects. (a) At a fixed state, known safe actions $(1,0,0)$ and $(0,0,1)$ certify the highlighted pairs. Combining these certificates gives $(1,0,1)$ without executing it. (b) On the four-agent path, safety requires equal actions and the reward is $\mathbf{1}\{a_1 = X\}$. With R communication rounds, the optimal safe reward $V_R^\star$ increases when X can reach agent 4 (Proposition H.1).

We study this problem in a cooperative Markov game with unknown reward, transition, and constraint models. Each constraint depends on the states and actions of a known group of agents. Training estimates the models centrally, and execution uses local states and messages exchanged over a communication graph. Agents coordinate before acting simultaneously. The comparator is the optimal safe protocol under the same communication restrictions.

Figure 1 reveals why training and communication are not interchangeable. Local certificates can establish an untried joint action's safety from previously collected evidence. Yet even with known models, agents cannot use observations that have not reached them before execution. Conversely, communication alone cannot resolve safety uncertainty left unidentified by the available data. The resulting design question is which safe observations and messages are sufficient to improve team performance without an unsafe trial. Answering it requires separate guarantees for learning which actions are safe and coordinating their execution. It also motivates jointly choosing safe measurements and communication schedules, beyond the fixed-resource conditions analyzed here.

Single-agent instantaneous-safe RL provides the statistical starting point [4–6]. The common reward and joint dynamics also admit a centralized MDP over joint actions. Factored MDP methods already exploit local structure and recover consistent actions through message passing [7, 8]. An unstructured joint-action table estimates repeated local configurations separately, while a full-state optimizer may use information unavailable to local agents. Local immediate rewards and constraints also do not guarantee efficient planning because future-value terms and exploration bonuses can enlarge the action scopes. Even a factor involving one action may require a distant agent's observation. Statistical complexity, planning complexity, and communication cost therefore require separate bounds.

SAFE-CMG-UCB combines local safety certificates with optimistic planning. A local tuple is admitted when either initial knowledge certifies it or a statistical upper bound establishes its safety. The same certificate can then be reused in every compatible joint action. The planner combines these certificates across overlapping scopes, retaining one shared action variable per agent, and selects an action using optimistic reward and transition estimates [9–11]. A known protocol safe at every modeled state remains available as a whole fallback, ensuring safe continuation after every action. A separate extension certifies continuation when the fallback is available only on a restricted domain.

Our first result establishes high-probability zero violation and a regret bound that separates reward and transition estimation from failure to certify optimal actions. Sublinear regret follows under exact communication-compatible planning, a positive margin for initially uncertified tuples of nonfallback comparator actions, and safely acquired observations that reduce uncertainty in the relevant constraint directions. To show why this information condition is essential, we construct two models that give identical observations under the fallback but disagree on whether a better action is safe. A learner that must remain safe in both models can then suffer linear regret. Thus a known safe policy need not provide the observations required to learn a better one.

Our structural results explain when coordination improves safe learning. We characterize what known constraint scopes and safe seed actions certify through local recombination, and bound tabular certification failures by local configuration counts. One example changes linear regret to constant regret when the specified learner reuses these local certificates. A second example has treewidth

one, yet insufficient communication prevents the team from attaining the full-state optimum. We then construct a junction-tree protocol that enforces one consistent action across overlapping scopes, including state acquisition and physical message routing. We bound communication rounds and the regret cost of finite-precision score messages. Finally, team regret bounds feasible unilateral-improvement gaps, clarifying the relation to constrained equilibrium objectives.

2 Related Work

Cumulative and instantaneous constraints. Constrained MDPs optimize reward subject to expected cumulative-cost budgets [12]. Regret analyses control budget violation [13, 14], and some methods guarantee that every selected policy satisfies its budget with high probability [15, 16]. These guarantees do not impose per-action thresholds. Expected instantaneous constraints on randomized action distributions likewise differ from actionwise safety [17]. Hard ramp and covering constraints also arise in online optimization with revealed feasibility information [18, 19]. Instantaneous-safe bandit and MDP methods are the closest statistical foundations [4–6, 20, 21]. Appendix B.1 compares their assumptions, safety geometry, and observation models with our $N = 1$ specialization.

Structured cooperation and safety. Factored planning and coordinated RL preserve local interactions and use variable elimination to recover consistent joint actions [7, 8]. We use this established planning machinery to combine initial and learned safety certificates, with explicit costs for certification and distributed implementation. Computational efficiency requires small induced width of the complete score-and-constraint graph. Shield decentralization starts from a given shield and sound environment abstraction [22]. Other approaches use constrained policy optimization or adaptive safety mechanisms [23, 24]. Differential-game approximations [25] and multiplayer reach–avoid analysis [26] address related control problems for specified models. Our analysis concerns learning unknown models while preserving instantaneous feasibility and controlling regret.

Equilibria and information structure. Constrained Markov games study feasible unilateral deviations and constrained equilibria [27–32]. Shared constraints determine which unilateral policy changes are feasible, so an equilibrium criterion requires a deviation class as well as a reward objective. In our common-reward protocol class, team regret bounds feasible unilateral-improvement gaps, while unilateral stability can coexist with a suboptimal team return (Corollary 5.7). Hierarchical-information cooperation [33] and Stackelberg learning [34, 35] impose distinct information or strategic response structures. Our elimination order controls computation and message organization while preserving the objective and feasible set. It introduces no strategic leader (Appendix B).

3 Problem Formulation

Notation. Write $[n] = \{1, \dots, n\}$, $[x]_+ = \max\{x, 0\}$, and $\Delta(\mathcal{X})$ for the probability simplex on a finite set $\mathcal{X}$. For $I \subseteq [N]$, $s_I = (s_i)_{i \in I}$ and $\mathbf{a}_I = (a_i)_{i \in I}$. For a positive definite matrix M, $\|x\|_M = (x^\top M x)^{1/2}$ is the induced norm. The symbol I_d denotes the d-dimensional identity. Matrix inequalities use the positive semidefinite order. Unmarked logarithms are natural.

3.1 Cooperative dynamics and hard safety

We consider $N \geq 1$ agents with finite nonempty local state spaces $\mathcal{S}_i$ and action sets $\mathcal{A}_i$. The nonempty global state space satisfies $\mathcal{S} \subseteq \prod_{i=1}^N \mathcal{S}_i$, and the joint action space is $\mathcal{A} = \prod_{i=1}^N \mathcal{A}_i$. The game has horizon $H \geq 1$, initial distribution $\rho \in \Delta(\mathcal{S})$, unknown transition kernels $\{P_h\}_{h=1}^H$, and unknown mean team rewards $r_h : \mathcal{S} \times \mathcal{A} \to [0, 1]$. Each of the $K \geq 1$ episodes starts with $s_1^k \sim \rho$, independently of the preceding history. At step h, the agents execute $\mathbf{a}_h^k \in \mathcal{A}$, and the next state is sampled as $s_{h+1}^k \sim P_h(\cdot \mid s_h^k, \mathbf{a}_h^k)$. We write $T = KH$ for the total interaction steps and $A_{\max} = \max_i |\mathcal{A}_i|$ for the maximum local action-space size.

There are $m \geq 1$ constraints. For each $\ell \in [m]$, a known constraint scope $I_\ell \subseteq [N]$ determines an unknown deterministic constraint function $g_{\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell})$. The safe joint-action set is

$$\mathcal{A}_h^{\mathrm{safe}}(s) = \{\mathbf{a} \in \mathcal{A} : g_{\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell}) \leq 0 \text{ for every } \ell \in [m]\}. \tag{1}$$

An agent in overlapping scopes must use the same action in every constraint involving it. Consequently, the intersection in (1) need not be a product of individual safe-action sets. Safety concerns the

latent constraint values, so measurement noise does not itself constitute a violation. Every executed action, including those used to collect training data, must satisfy these constraints.

3.2 Communication-compatible protocols

At step h, agent i observes its local state s_i. Before executing a joint action, agents communicate over an undirected graph $G_{\mathrm{comm}} = ([N], E_{\mathrm{comm}})$ for a given integer budget $R_h \geq 0$ of synchronous rounds. A step-h protocol consists of message maps and action maps,

$$z^t_{i\to j} = \mu^t_{i\to j,h}(s_i, \mathsf{M}^{t-1}_i), \qquad a_i = \pi_{i,h}(s_i, \mathsf{M}^{R_h}_i), \tag{2}$$

where $z^t_{i\to j}$ is the message from i to j in round $t \in [R_h]$, M^t_i is agent i's transcript through round t, M^0_i is empty, $a_i \in \mathcal{A}_i$, and $(i,j) \in E_{\mathrm{comm}}$. Centralized training installs local model and message programs before each episode. Agents evaluate them using local observations and received messages. All agents act simultaneously after communication completes. Communication is reliable, the state remains fixed during these rounds, and messages and delays do not enter the reward or horizon. The exact construction permits an arbitrary finite vector per directed edge per round. Section 4.3 accounts separately for message volume and precision.

Let $\mathfrak{P}$ be the class of deterministic Markov protocols. Unless otherwise specified, it contains all local message and action maps satisfying the graph, message alphabet, and round budgets. A more restricted class must still satisfy Assumption 3.3. Tie rules are fixed before observing the current state. We write $\Pi_h(s)$ for the joint action induced by running the distributed protocol at s, although each agent uses only its own observation and received messages. A protocol is safe if every constraint holds almost surely along its trajectories from ρ. Its value is

$$V^\Pi_h(s) = \mathbb{E}_\Pi\left[\sum\nolimits_{t=h}^{H} r_t(s_t, \mathbf{a}_t) \mid s_h = s\right], \qquad V^\Pi_1(\rho) = \mathbb{E}_{s\sim\rho}[V^\Pi_1(s)]. \tag{3}$$

The expectation starts from state s at step h, including states unreachable from ρ. For any state function v, write $v(\rho) = \sum_s \rho(s)v(s)$. Let $\mathfrak{P}_{\mathrm{safe}}$ denote the safe protocols and choose an optimal safe comparator $\Pi^\star \in \arg\max_{\Pi\in\mathfrak{P}_{\mathrm{safe}}} V^\Pi_1(\rho)$. The interaction history before episode k is $\mathcal{H}_{k-1}$, and Π^k is chosen using this history. Cumulative regret and positive constraint violation are

$$\mathrm{Reg}(K) = \sum\nolimits_{k=1}^{K}\left[V^{\Pi^\star}_1(\rho) - V^{\Pi^k}_1(\rho)\right], \qquad \mathrm{Viol}(K) = \sum\nolimits_{k,h,\ell}[g_{\ell,h}(s^k_{I_{\ell,h}}, \mathbf{a}^k_{I_{\ell,h}})]_+. \tag{4}$$

For a failure probability $\delta \in (0,1)$, the goal is $\mathrm{Viol}(K) = 0$ with probability at least $1-\delta$ and sublinear regret under the stated information conditions.

3.3 Statistical model and observations

Let $\mathcal{E}_r$ be a family of reward scopes $e \subseteq [N]$, and let $r_{e,h}$ be the mean of reward factor e. Reward and constraint functions have local features. Transitions follow a linear-mixture model that may couple all agents. These known features permit sharing observations across inputs [10, 11], as specified by

$$r_h(s,\mathbf{a}) = \sum\nolimits_{e\in\mathcal{E}_r} r_{e,h}(s_e, \mathbf{a}_e), \quad r_{e,h}(s_e,\mathbf{a}_e) = \langle \phi^r_{e,h}(s_e,\mathbf{a}_e), \theta^r_{e,h}\rangle, \tag{5}$$

$$g_{\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell}) = \langle \phi^g_{\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell}), \theta^g_{\ell,h}\rangle, \tag{6}$$

$$P_hV(s,\mathbf{a}) = \langle \psi^V_h(s,\mathbf{a}), \theta^P_h\rangle, \qquad \psi^V_h(s,\mathbf{a}) = \sum\nolimits_{s'\in\mathcal{S}} \phi^P_h(s' \mid s,\mathbf{a})V(s'). \tag{7}$$

Here $P_hV(s,\mathbf{a}) := \sum_{s'} P_h(s' \mid s,\mathbf{a})V(s')$ is the expected next-state value. All feature maps and scopes are known, while the parameters are unknown. The transition parameter belongs to a known set Θ^P_h for which $P_\theta(s' \mid s,\mathbf{a}) = \langle \phi^P_h(s' \mid s,\mathbf{a}), \theta\rangle$ is a probability kernel. The same θ^P_h applies to every V.

Assumption 3.1 (Bounded linear model and feedback)**.** Local features have Euclidean norm at most one, and $\|\theta^r_{e,h}\|_2 \leq B_r$, $\|\theta^g_{\ell,h}\|_2 \leq B_g$, and $\|\theta^P_h\|_2 \leq B_P$, for given nonnegative bounds. Also $\|\psi^V_h(s,\mathbf{a})\|_2 \leq \|V\|_\infty$ for every $V : \mathcal{S} \to \mathbb{R}$. At each executed step the learner observes the next state, each local reward-factor measurement $Y_{r,e,h}$, and each constraint measurement $Y_{g,\ell,h}$. Their conditional means are the corresponding local reward and constraint values. The centered noises are conditionally R_r- and R_g-sub-Gaussian for given nonnegative noise bounds. Any auxiliary constraint observation has response $Y = \langle x, \theta^g_{\ell,h}\rangle + \varepsilon$, where x is a predictable feature with $\|x\|_2 \leq 1$ and ε is conditionally R_g-sub-Gaussian noise. It is obtained without executing an uncertified action.

Let $F = |\mathcal{E}_r|$ count reward factors and let $d_{r,e} = \dim(\phi^r_{e,h})$, $d_g = \dim(\phi^g_{\ell,h})$, and $d_P = \dim(\phi^P_h)$ be the reward, constraint, and transition feature dimensions. Zero-padding gives positive stage-independent dimensions and a common d_g. The total reward dimension is $D_r = \sum_e d_{r,e}$. Separate reward regressions require factor observations. Appendix G.1 gives an aggregate-reward variant. Let $n_g \geq K$ bound all observations per (ℓ, h) through episode K. Auxiliary acquisition costs are separate from environment regret.

Assumption 3.2 (Available safe fallback)**.** A known protocol $\Pi^0 \in \mathfrak{P}$ induces $\mathbf{a}^0_h(s)$ satisfying $g_{\ell,h}(s_{I_\ell}, \mathbf{a}^0_{I_{\ell,h}}(s)) \leq 0$ for every $s \in \mathcal{S}$, $h \in [H]$, and $\ell \in [m]$.

The learner may additionally use fixed, sound local certificates

$$\mathcal{Z}_\ell = \{(s_{I_\ell}, \mathbf{a}_{I_\ell}) : (s, \mathbf{a}) \in \mathcal{S} \times \mathcal{A}\}, \quad \mathcal{Z}^0_{\ell,h} \subseteq \{z \in \mathcal{Z}_\ell : g_{\ell,h}(z) \leq 0\}. \tag{8}$$

Here $\mathcal{Z}_\ell$ contains attainable local tuples and $\mathcal{Z}^0_{\ell,h}$ is its initially certified subset, possibly empty. Projecting known safe joint state–action pairs, including fallback pairs, onto the constraint scope provides valid certificates. Each certificate applies to every global configuration containing that tuple, enabling the recombination in Figure 1(a), including boundary-safe tuples.

Exploration can reach states outside the fallback's own trajectories. Assumption 3.2 therefore covers every modeled state without requiring a strict margin. Appendix F studies continuation certificates when this global fallback condition is unavailable.

Assumption 3.3 (Exact, communication-compatible planning)**.** For each episode, the selected maximizers of the unclipped score (16) over (15), using the specified tie rule, are realizable jointly for all states by one protocol $\Pi^k \in \mathfrak{P}$, including the comparison with Π^0. The planner returns this protocol and its exact clipped values (17).

Separate statewise optimizers need not define one admissible policy when agents cannot distinguish states requiring different actions. Section 4.3 provides a construction with evaluable factors, bounded induced width, and sufficient communication for state acquisition and message passing. Linear realizability does not ensure these conditions.

3.4 Examples and structural limits

In discretized robot coordination, minimum next-position separation couples robot controls, while receiver interference limits couple transmitter powers. Chain interactions can have small planning width, but all-to-all pairwise collision checks induce a complete action graph and network-wide interference can create a global constraint. Efficient planning also requires suitable Bellman and bonus scopes. Appendix E.5 gives explicit constraints, sparse and dense variants, and their feature, feedback, and prediction-error requirements.

A communication-dependent safe optimum. Consider a known-model, one-step game on the communication path $1 - \cdots - N$. Only agent 1 observes a fair bit X. The reward is $\mathbf{1}\{a_1 = X\}$, and pairwise constraints force all binary actions to agree. The all-zero fallback is safe with margin one. Despite action-graph treewidth one,

$$V^\star_{\text{full}} = 1, \qquad V^\star_R = 1/2 \ (R < N - 1), \qquad V^\star_R = 1 \ (R \geq N - 1). \tag{9}$$

Here $V^\star_{\text{full}}$ is the full-state optimum and $V^\star_R$ is the optimum over safe R-round protocols. Agent N must agree with agent 1 without knowing X when $R < N - 1$, which proves the value gap. Broadcasting X attains value one when $R \geq N - 1$. The full-state optimizer is implementable only at or above this threshold, and Assumption 3.3 fails below it. Figure 1(b) shows the four-agent instance, and Proposition H.1 gives the full proof in Appendix H.

4 Safe Learning and Distributed Action Coordination

SAFE-CMG-UCB uses optimistic reward and transition estimates to encourage exploration, while upper confidence bounds on constraints exclude actions whose safety is unresolved. Each episode fits models, plans backward, and executes a fixed distributed policy. Algorithm 1 specifies these phases, Figure 2 relates them to performance and safety, and Figure 3 details the protocol.

4.1 Predictable regression and confidence bounds

Before episode k, identity-regularized least squares estimates each local reward and constraint model and one transition model per step. Let $\Lambda^r_{k,e,h}, \Lambda^g_{k,\ell,h}, \Lambda^P_{k,h}$ denote the design matrices, $\widehat{\theta}^r_{k,e,h}, \widehat{\theta}^g_{k,\ell,h}, \widehat{\theta}^P_{k,h}$ the parameter estimates, and $\beta^r_{k,e,h}, \beta^g_{k,\ell,h}, \beta^P_{k,h}$ the confidence radii. Appendices A.1 and A.2 give all estimators and parameters. The designs record observed feature directions and determine prediction uncertainty.

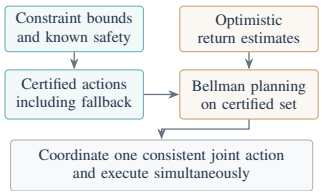


Figure 2: On the confidence event, consistent certified actions ensure safety.

Allocating failure probability $\alpha = \delta/[2H(F+m+1)]$ to each regression, self-normalized concentration [36] yields a simultaneous parameter-confidence event $\mathcal{E}$ with probability at least $1-\delta/2$ (Appendix C.1). On $\mathcal{E}$, the ellipsoids bound predictions uniformly over states, actions, and value functions. Appendix C.6 explains why this construction needs no separate value-function covering argument.

Transition regression predicts next-state value. Its normalized feature $x^\tau_{P,h}$ and response $Y^\tau_{P,h}$ retain the value function fixed before the sample was collected,

$$x^\tau_{P,h} = \psi_h^{V_{\tau,h+1}}(s^\tau_h, \mathbf{a}^\tau_h)/H, \qquad Y^\tau_{P,h} = V_{\tau,h+1}(s^\tau_{h+1})/H. \tag{10}$$

Apply (31) for $\tau < k$ to obtain $\Lambda^P_{k,h}$ and $\widehat{\theta}^P_{k,h}$. The feature is measurable before s^τ_{h+1} is observed. Clipping keeps values in $[0,H]$ on every history, and division by H bounds both the response and feature norm by one. Substituting later value estimates need not preserve this predictability.

The estimated mean reward and expected next-state value are $\widehat{r}_{k,h} = \sum_e \langle \phi^r_{e,h}, \widehat{\theta}^r_{k,e,h}\rangle$ and $\widehat{P}_{k,h}V = \langle \psi^V_h, \widehat{\theta}^P_{k,h}\rangle$, with the features' state–action arguments. The estimate $\widehat{P}_{k,h}$ need not be a probability kernel. Define the reward bonus b^r, transition bonus b^P, and constraint upper bound U^g by

$$b^r_{k,h}(s,\mathbf{a}) = \sum\nolimits_{e\in\mathcal{E}_r} \beta^r_{k,e,h} \|\phi^r_{e,h}(s_e,\mathbf{a}_e)\|_{(\Lambda^r_{k,e,h})^{-1}}, \tag{11}$$

$$b^P_{k,h}(s,\mathbf{a};V) = H\beta^P_{k,h}\|\psi^V_h(s,\mathbf{a})/H\|_{(\Lambda^P_{k,h})^{-1}}, \tag{12}$$

$$U^g_{k,\ell,h}(s_{I_\ell},\mathbf{a}_{I_\ell}) = \langle \phi^g_{\ell,h}(s_{I_\ell},\mathbf{a}_{I_\ell}), \widehat{\theta}^g_{k,\ell,h}\rangle + \beta^g_{k,\ell,h}\|\phi^g_{\ell,h}(s_{I_\ell},\mathbf{a}_{I_\ell})\|_{(\Lambda^g_{k,\ell,h})^{-1}}. \tag{13}$$

On $\mathcal{E}$, $U^g_{k,\ell,h} \ge g_{\ell,h}$, $\widehat{r}_{k,h} + b^r_{k,h} \ge r_h$, and $\widehat{P}_{k,h}V + b^P_{k,h}(\cdot;V) \ge P_hV$. Reward and transition uncertainty enlarge optimistic return bounds, whereas constraint uncertainty makes safety certification more conservative. Only certified candidates are executed.

4.2 Certified action sets and optimistic Bellman backups

The local state–action tuple is $z_\ell(s,\mathbf{a}) = (s_{I_\ell}, \mathbf{a}_{I_\ell})$. Each constraint is certified by initial knowledge or its upper confidence bound. Requiring a valid certificate in every scope gives the immediate certified set $\widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s)$. Retaining the whole fallback yields the candidate set $\widehat{\mathcal{A}}_{k,h}(s)$,

$$\widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s) = \{\mathbf{a} :\ z_\ell(s,\mathbf{a}) \in \mathcal{Z}^0_{\ell,h} \text{ or } U^g_{k,\ell,h}(z_\ell(s,\mathbf{a})) \le 0,\ \text{for every } \ell\}, \tag{14}$$

$$\widehat{\mathcal{A}}_{k,h}(s) = \widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s) \cup \{\mathbf{a}^0_h(s)\}. \tag{15}$$

Every candidate is safe on $\mathcal{E}$, and the fallback makes the set nonempty. Initially certified boundary tuples can be combined with learned certificates. Appendix J.2 gives regret K when the fallback is retained only as an indivisible action and $\min\{K,3\}$ when its local certificates are reused, without auxiliary observations. Initialize $V_{k,H+1} = 0$. The unclipped optimistic score $\widetilde{Q}_{k,h}$, clipped action value $Q_{k,h}$, and state value $V_{k,h}$ satisfy the backward recursion

$$\widetilde{Q}_{k,h}(s,\mathbf{a}) = \widehat{r}_{k,h}(s,\mathbf{a}) + b^r_{k,h}(s,\mathbf{a}) + \widehat{P}_{k,h}V_{k,h+1}(s,\mathbf{a}) + b^P_{k,h}(s,\mathbf{a};V_{k,h+1}), \tag{16}$$

$$Q_{k,h}(s,\mathbf{a}) = \mathrm{clip}_{[0,H-h+1]}\left(\widetilde{Q}_{k,h}(s,\mathbf{a})\right), \quad V_{k,h}(s) = \max\nolimits_{\mathbf{a}\in\widehat{\mathcal{A}}_{k,h}(s)} Q_{k,h}(s,\mathbf{a}). \tag{17}$$

Algorithm 1 SAFE-CMG-UCB

Input: State and action sets, known features and scopes, bounds B_r, B_g, B_P, R_r, R_g, protocol class $\mathfrak{P}$ with its graph, alphabet, round budgets and tie rule, safe fallback Π^0, optional local certificates $\mathcal{Z}^0_{\ell,h}$, H, K and $\delta \in (0,1)$.

Initialize the regression data sets to be empty.
For $k = 1, \dots, K$:
  Estimate: fit the models and confidence sets from pre-episode data using (31)–(33).
  Set $V_{k,H+1}(s) = 0$ for every $s \in \mathcal{S}$.
  For $h = H, \dots, 1$:
    Certify: combine local certificates and the whole fallback in (15).
    Plan: compute the optimistic scores (16) and values (17). Compile the factor evaluators, routes, and protocol maps (2) for all states, including fallback selection.
  End for
  For $h = 1, \dots, H$:
    Coordinate and act: instantiate factors from local observations and messages. Agree on the certified assignment or whole fallback, then execute $\Pi^k_h(s^k_h)$ simultaneously.
    Observe: store factor rewards, constraint measurements, and the historical-value transition sample (10).
  End for
  Incorporate any safely obtained auxiliary constraint observations permitted by the feedback model.
End for

Here $\mathrm{clip}_{[0,b]}(x) = \max\{0, \min\{b, x\}\}$. Select a maximizer of the unclipped score, preferring the whole fallback on ties and otherwise using fixed action and bag orders. Monotone clipping preserves this maximizer, and Assumption 3.3 makes these choices one admissible protocol.

The globally available fallback ensures safe continuation, so the main algorithm requires only immediate certificates. Replacing this assumption by backward viability tests requires the additional transition-support conditions in Appendix F.

4.3 Exact coordination, communication, and complexity

Suppose the complete score is a known sum of local score factors $q_{k,h,f}$, indexed by $\mathcal{F}^Q_{k,h}$. The score decomposition and constrained planning problem are

$$\begin{aligned} &\widetilde{Q}_{k,h}(s, \mathbf{a}) = \sum\nolimits_{f \in \mathcal{F}^Q_{k,h}} q_{k,h,f}(s_{J_f}, \mathbf{a}_{I_f}), \\ &\max\nolimits_{\mathbf{a} \in \mathcal{A}} \sum\nolimits_f q_{k,h,f}(s_{J_f}, \mathbf{a}_{I_f}) \quad \text{s.t.} \quad z_\ell \in \mathcal{Z}^0_{\ell,h} \text{ or } U^g_{k,\ell,h}(z_\ell) \le 0 \ \ (\ell \in [m]). \end{aligned} \tag{18}$$

The second expression optimizes over immediate certified actions. Comparing its optimum with the whole fallback implements (15), even if the immediate set is empty. All transition-value and bonus terms are included. Action scopes I_f determine elimination width, while state scopes J_f determine the observations required for factor evaluation.

Exact variable elimination or junction-tree computation solves (18) [7, 8, 37]. Upward messages summarize conditional scores and aggregate the fallback score separately. The root selects the certified optimizer or whole fallback. Downward assignments then fix one value for every shared action variable, ensuring consistency across overlapping constraints. Messages follow routes on G_{comm}, and all agents act after delivery (Appendix E).

Let $\mathcal{F}$ include all score and safety factors, $r_{\max}$ their maximum action-scope size, and ω the induced width of the chosen elimination order on their primal graph. Thus ω is the largest number of remaining neighbors of an eliminated variable after fill edges are added. Constructing the tables takes $O(|\mathcal{F}| A^{r_{\max}}_{\max})$ factor evaluations. For a reduced junction tree $\mathcal{T}$ with bag collection $\mathcal{B}$, exact optimization of these tables requires

$$O\big((|\mathcal{F}| + |\mathcal{B}|) A^{\omega+1}_{\max} + N\big) \tag{19}$$

arithmetic operations per state, plus the fallback evaluation. Reduced junction-tree separator tables have $O(A^{\omega}_{\max})$ scalar entries. The two passes take $O(\mathrm{diam}(\mathcal{T}))$ logical rounds for tree diameter $\mathrm{diam}(\mathcal{T})$, plus routing, state-acquisition, and congestion costs on G_{comm}.

Proposition I.1 compiles this computation into one protocol over all states. Let R_0 be the fallback's round cost, r_{in} the largest required input-routing distance, r_{out} the largest output-routing distance, and D_G the largest sum of physical routing distances on a root-to-leaf junction-tree path. With fixed bag hosts and routes, sufficient rounds are

$$R \geq R_0 + r_{\mathrm{in}} + 2D_G + r_{\mathrm{out}}. \tag{20}$$

This bound includes state acquisition, fallback comparison, assignment, and delivery under the vector-message model. Proposition I.2 also bounds routed bit traffic. Rounding only score factors with quantization step $\zeta > 0$, while keeping safety indicators exact, preserves feasibility and adds at most $|\mathcal{F}^Q_{k,h}|\zeta$ optimization error.

Runtime factor instantiation and message passing evaluate the estimates, certificates, values, and protocol maps fixed before the episode. Proposition E.1 proves equivalence with precomputation under the same tie rule. Tabular planning requires $H|\mathcal{S}|$ state–step evaluations, with separate factor evaluation, storage, and distribution costs. A host may need global state inputs. Dense pairwise scopes or a global Bellman factor can give $\omega = N - 1$ and exponential action cost.

5 Theoretical Guarantees

The largest confidence radii are $\beta^r = \max_{k,e,h} \beta^r_{k,e,h}$, $\beta^P = \max_{k,h} \beta^P_{k,h}$, and $\beta^g = \max_{k,\ell,h} \beta^g_{k,\ell,h}$, with maxima over the first K episodes. The factor $\sqrt{F}$ below comes from summing confidence widths of separately estimated reward factors. Equation (46) gives deterministic bounds on these maxima. The notation $\widetilde{O}$ suppresses logarithmic factors in $K, n_g, 1/\delta$.

Theorem 5.1 (Zero instantaneous violation)**.** *Under Assumptions 3.1 and 3.2, an implementation of* SAFE-CMG-UCB *that executes only actions in* (15) *satisfies every instantaneous constraint with probability at least* $1 - \delta$*. Hence* $\mathrm{Viol}(K) = 0$ *on this event. This conclusion does not require a positive comparator margin or a certification-coverage condition.*

The guarantee holds even when uncertainty prevents certification of the optimal protocol. *Proof sketch.* On the simultaneous confidence event, every statistical upper bound dominates its true constraint. Known local certificates and the all-state fallback cover the other candidates. Every executed joint action is therefore safe. The full proof is in Appendix C.2.

Performance also depends on which safe actions can be certified. Let

$$\mathcal{X}^\star_h := \{(s, \mathbf{a}) : \mathbb{P}_{\Pi^\star}(s_h = s, \mathbf{a}_h = \mathbf{a}) > 0\}. \tag{21}$$

This is the support of the comparator's step-h state–action distribution under the true kernels and initial law ρ. Let χ_k indicate an uncertified action in the comparator's reachable support in episode k, and let B_K count such episodes,

$$\chi_k := \mathbf{1}\Big\{\exists h \in [H],\ (s, \mathbf{a}) \in \mathcal{X}^\star_h : \mathbf{a} \notin \widehat{\mathcal{A}}_{k,h}(s)\Big\}, \qquad B_K := \sum\nolimits_{k=1}^{K} \chi_k. \tag{22}$$

Neither $\Pi^\star$, χ_k, nor B_K is an input to the learner.

Theorem 5.2 (Regret with an explicit certification term)**.** *Under Assumptions 3.1–3.3, zero violation and the following regret bound hold jointly with probability at least* $1 - \delta$*,*

$$\begin{aligned}\mathrm{Reg}(K) \leq\ & 2\beta^r \sqrt{2FHD_r T \log(1 + K)} + 2H\beta^P \sqrt{2Hd_P T \log(1 + K/d_P)} \\ & + HB_K + H\sqrt{2T \log(2/\delta)}.\end{aligned} \tag{23}$$

The first two terms measure reward and transition uncertainty, while HB_K measures the cost of excluding the comparator. The final term controls trajectory-sampling error. The bound is sublinear when $B_K = o(K)$ and its estimation terms are sublinear. *Proof sketch.* When $\chi_k = 0$, backward induction establishes comparator optimism. Otherwise the lost value is at most H. Telescoping the executed policy's Bellman errors reduces the remaining regret to confidence widths and trajectory-sampling errors. Stepwise elliptical-potential bounds control the former, and martingale concentration controls the latter. The full proof is in Appendix C.3.

Assumption 5.3 (Comparator margin and certification coverage)**.** There exists $\Delta_{\mathrm{safe}} > 0$ such that, for every $h \in [H]$, $\ell \in [m]$, and $(s, \mathbf{a}) \in \mathcal{X}_h^\star$ with $\mathbf{a} \neq \mathbf{a}_h^0(s)$,

$$z_\ell(s, \mathbf{a}) \notin \mathcal{Z}_{\ell,h}^0 \quad \Longrightarrow \quad g_{\ell,h}(z_\ell(s, \mathbf{a})) \leq -\Delta_{\mathrm{safe}}. \tag{24}$$

Initially certified tuples may lie on the safety boundary. On the confidence event, for every episode with $\chi_k = 1$, there is a witness $(h, \ell, s, \mathbf{a})$ satisfying $(s, \mathbf{a}) \in \mathcal{X}_h^\star$, $\mathbf{a} \notin \widehat{\mathcal{A}}_{k,h}(s)$, $z_\ell(s, \mathbf{a}) \notin \mathcal{Z}_{\ell,h}^0$, and $U_{k,\ell,h}^g(z_\ell(s, \mathbf{a})) > 0$ such that, for $x = \phi_{\ell,h}^g(s_{I_\ell}, \mathbf{a}_{I_\ell})$,

$$\Lambda_{k+1,\ell,h}^g - \Lambda_{k,\ell,h}^g \succeq xx^\top. \tag{25}$$

All observations contributing to this increase are acquired without violating the hard constraints. This condition does not authorize executing an uncertified candidate. The budget n_g counts all measurements per (ℓ, h) through episode K, including safe auxiliary feedback.

In each episode with an excluded comparator action, condition (25) requires safely collected data to provide the specified covariance increase for at least one failed local test. Executed-action feedback or additional safety measurements may provide this information, with auxiliary acquisition costs accounted for separately. Proposition C.8 gives a trajectory-feedback instance with linear regret despite a safe fallback, positive margin, and sparse scopes. The margin and coverage witnesses are analytical conditions, not algorithm inputs.

Corollary 5.4 (Sublinear regret under certification coverage)**.** *Under the assumptions of Theorem 5.2 and Assumption 5.3, on the same event,*

$$B_K \leq 8Hmd_g(\beta^g/\Delta_{\mathrm{safe}})^2 \log(1 + n_g/d_g). \tag{26}$$

Substituting this certification count into Theorem 5.2 gives

$$\mathrm{Reg}(K) \leq \widetilde{O}\Big(\beta^r \sqrt{FHD_rT} + H\beta^P \sqrt{HdPT} + H^2 md_g(\beta^g/\Delta_{\mathrm{safe}})^2 + H\sqrt{T}\Big). \tag{27}$$

For fixed H, F, m, dimensions, parameter and noise bounds, $\Delta_{\mathrm{safe}} > 0$, and $\delta \in (0, 1)$, if n_g is polynomial in K, the confidence radii are $O(\sqrt{\log K})$ and the regret is sublinear in K.

The margin sets the confidence-width threshold, while coverage limits how often certification can fail. *Proof sketch.* For each episode counted by B_K, assign one failed test based on Assumption 5.3. Its margin implies a confidence half-width exceeding $\Delta_{\mathrm{safe}}/2$, while coverage provides a corresponding observed design increment. Summing the resulting increases in $\log\det\Lambda$ proves (26). The full proof is in Appendix C.4. Certified approximate planning with state-uniform score error $\varepsilon_{k,h}$ increases regret by at most $\sum_{k,h} \varepsilon_{k,h}$, provided every output remains feasible (Appendix C.5).

Single-agent specialization. For $N = 1$, communication disappears and joint tuples become ordinary state–action pairs. With one reward factor, substituting $F = 1$ and $D_r = d_r$ into (27) gives the rate in Appendix B.1. This specialization retains the all-state fallback, unknown-tuple margin, and feedback coverage. No sharper rate than existing instantaneous-safe MDP results is claimed [4–6]. The appendix compares these assumptions and distinguishes actionwise, expected instantaneous, and cumulative constraints.

Theorem 5.5 (Necessity lower bounds)**.** *The following statements hold.*

1. *For unstructured one-state, one-step cooperative games with N agents, $A \geq 2$ actions per agent, and all joint actions safe, every algorithm has expected regret at least $c\sqrt{A^N K}$ on some instance when $K \geq A^N$, with $c = 1/(16\sqrt{2})$.*

2. *Without initial safety information or safety feedback before the first action, some instance has first-action violation probability at least $1/2$. A uniform zero-violation guarantee with failure probability $\delta < 1/2$ is therefore impossible.*

3. *Let $\gamma > 0$ and $0 < \delta \leq 1/4$. Given n independent observations $Y_1, \ldots, Y_n \sim \mathcal{N}(\theta, 1)$ of a candidate constraint with mean $\theta \in \{-\gamma, +\gamma\}$, any certification rule that declares the safe candidate safe with probability at least $1 - \delta$ and declares the unsafe candidate safe with probability at most δ requires $n \geq (2\gamma^2)^{-1} \mathrm{kl}(1 - \delta, \delta)$. Here $\mathrm{kl}(p, q) = p \log(p/q) + (1 - p) \log((1 - p)/(1 - q))$.*

The three bounds distinguish representation complexity, initial safety information, and certification precision. The first concerns unstructured rewards, so retaining reward structure can change the statistical complexity. The third quantifies the cost of distinguishing near-boundary safe and unsafe actions and requires both soundness and successful certification. *Proof sketch.* An A^N-armed Bernoulli bandit gives the first bound by change of measure. The second follows from indistinguishable initial safe-action assignments. The third uses Gaussian binary testing. Full proofs appear in Appendix D.

5.1 Local tabular certification and feasible deviations

For tabular local constraints, use the sample-mean upper confidence bound $U^{g,\mathrm{tab}}_{k,\ell,h}(z)$ defined in Appendix A.3, retaining the other algorithm components. Let $M_\ell = |\mathcal{Z}_\ell|$ count local configurations and $L_\ell = \log(2M_\ell n_g/\alpha)$ be the logarithmic confidence factor, with a given integer observation budget $n_g \geq K$ per (ℓ, h). The count of an entry is the number of its observations before episode k. The sufficient certification count q_ℓ below is used only in the analysis.

Proposition 5.6 (Local tabular certification)**.** *Assume the model, fallback, planning, and unknown-tuple margin conditions. Suppose each episode with $\chi_k = 1$ has an unavailable comparator action $\mathbf{a} \notin \widehat{\mathcal{A}}_{k,h}(s)$ and a failed test $U^{g,\mathrm{tab}}_{k,\ell,h}(z) > 0$ at $z = z_\ell(s, \mathbf{a}) \notin \mathcal{Z}^0_{\ell,h}$, whose count increases by at least one through safely acquired observations in that episode. Set $q_\ell = \max\{1, \lceil 8R_g^2 L_\ell/\Delta^2_{\mathrm{safe}}\rceil\}$. With probability at least $1-\delta$, zero violation and Theorem 5.2's regret bound hold jointly, with*

$$B_K \leq \sum\nolimits_{h,\ell} |\mathcal{Z}_\ell \setminus \mathcal{Z}^0_{\ell,h}| q_\ell \leq H \sum\nolimits_{\ell=1}^{m} M_\ell q_\ell. \tag{28}$$

Each local estimate is reused by all joint actions containing the same tuple. A one-state binary pairwise chain has $4H(N-1)$ local entries, versus $H(N-1)2^N$ full constraint entries. This gain is also available to centralized planners that retain the same structure. *Proof sketch.* A failed comparator test has count below q_ℓ, and coverage increases one such count. Each initially uncertified entry can therefore be charged at most q_ℓ times. The full proof is in Appendix J.3.

For safe Π, let $\mathfrak{D}_i(\Pi)$ comprise replacements of agent i's message and action maps, with the other maps fixed, for which the entire resulting protocol remains in $\mathfrak{P}_{\mathrm{safe}}$. Include the identity and define the feasible unilateral-improvement gap

$$\mathrm{Gap}_i(\Pi) = \sup\nolimits_{\widetilde{\Pi}\in\mathfrak{D}_i(\Pi)} [V_1^{\widetilde{\Pi}}(\rho) - V_1^{\Pi}(\rho)]. \tag{29}$$

Corollary 5.7 (Feasible unilateral improvements)**.** *For every safe Π, $0 \leq \max_i \mathrm{Gap}_i(\Pi) \leq V_1^{\Pi^\star}(\rho) - V_1^{\Pi}(\rho)$. On the simultaneous confidence and regret event of Theorem 5.2,*

$$\sum\nolimits_{k=1}^{K} \max_i \mathrm{Gap}_i(\Pi^k) \leq \mathrm{Reg}(K). \tag{30}$$

Sublinear team regret gives a vanishing average maximum feasible-deviation gap. This implication is strict. With no messages, safe pairs $\{00, 11\}$ and reward $(a_1 + a_2)/2$ make 00 unilaterally stable but team-suboptimal by one. The result concerns deviations from ρ. General-sum and off-path equilibrium guarantees require different analyses. *Proof sketch.* The identity deviation gives a nonnegative gap. Every feasible unilateral deviation remains in the comparator's safe protocol class, so optimality of $\Pi^\star$ bounds its gain by team suboptimality. Taking the supremum, maximizing over agents, and summing proves the result. The full proof is in Appendix J.4.

6 Conclusion

SAFE-CMG-UCB combines reusable local certificates, optimistic Bellman planning, and consistent joint-action coordination for safe cooperative learning under coupled instantaneous constraints and limited communication. With an all-state safe fallback, it guarantees high-probability zero violation. The regret bound separates estimation and certification costs and yields sublinear regret under the stated safety-margin, information-coverage, and implementability conditions. Structural bounds quantify local certification and coordination costs, while impossibility results identify distinct information and communication barriers to learning and attaining the full-state safe optimum. Two questions remain. Can adaptive safe measurements achieve coverage with minimal certification delay? What safe value and regret are attainable when communication prevents exact planning?

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A Notation and Algorithm Parameters

Table 1: Notation used in the main results and proofs.

Symbol	Meaning
N, H, K, $T = KH$	Agents, steps per episode, episodes, and total interactions.
$\mathcal{S}$, $\mathcal{S}_i$, $\mathcal{A}_i$, $\mathcal{A}$	Global state space, local state space, individual action space, and joint-action space.
$\mathrm{proj}_J\,\mathcal{S}$, $\mathcal{A}_I$	Local state tuples attainable on scope J and the action product $\prod_{i\in I}\mathcal{A}_i$.
s_I, $\mathbf{a}_I$, I_ℓ	State and action restrictions to a set of agents, and the scope of constraint ℓ.
r_h, P_h, ρ	Team reward mean, transition kernel, and initial-state law.
m, $g_{\ell,h}$, $\mathcal{A}_h^{\mathrm{safe}}(s)$	Constraint count, constraint function, and true safe actions.
G_{comm}, $\mu^t_{i\to j,h}$, $\pi_{i,h}$	Physical graph, message rule, and local action rule.
$\mathcal{N}_i^{\mathrm{comm}}$	Physical communication neighbors of agent i.
$z^t_{i\to j}$, M^t_i	Round-t message and agent i's transcript, initially empty.
$\mathfrak{P}$, $\mathfrak{P}_{\mathrm{safe}}$	Admissible protocol class and its safe subset.
Π^0, Π^k, $\Pi^\star$	Known safe fallback, executed protocol in episode k, and optimal safe admissible protocol.
$\mathcal{H}_{k-1}$	Feedback and training information before episode k.
$\mathcal{H}^{\mathrm{pre}}_{\tau,h}$	History just before the step-h transition in episode τ, including the chosen state–action pair.
V_h^Π, $V_{k,h}$	Value of Π and optimistic value for episode k.
$\mathcal{E}_r$, $F = \lvert\mathcal{E}_r\rvert$, $d_{r,e}$	Reward scopes, number of factors, and dimension of factor e.
$D_r = \sum_e d_{r,e}$, d_g, d_P	Total reward dimension, padded constraint dimension, and padded transition dimension.
$\phi^r_{e,h}$, $\phi^g_{\ell,h}$, ϕ^P_h	Known reward, constraint, and transition-kernel features.
$\psi^V_h = \sum_{s'}\phi^P_h(s' \mid \cdot)V(s')$	Transition feature for the expectation of V.
$\theta^r_{e,h}$, $\theta^g_{\ell,h}$, θ^P_h	Unknown model parameters. Hats indicate ridge estimates.
B_r, B_g, B_P, R_r, R_g, $R_P = 1$	Parameter norm bounds and sub-Gaussian noise scales. $R_P = 1$ bounds the normalized transition noise.
$\mathcal{D}_k$, $\mathcal{D}^g_{k,\ell,h}$	Indices of individual observations available before episode k in a generic regression and a constraint regression.
$x^\tau_{r,e,h}$, $Y^\tau_{r,e,h}$, $x^\tau_{P,h}$, $Y^\tau_{P,h}$	Reward feature and factor measurement, followed by the historical-value transition feature and target. Only the transition pair is normalized by H.
$\Lambda^r_{k,e,h}$, $\Lambda^g_{k,\ell,h}$, $\Lambda^P_{k,h}$	Identity-regularized design matrices. Transition features and targets are normalized by H.
$\widehat{r}_{k,h}$, $\widehat{P}_{k,h}V$	Estimated mean reward and estimated next-value expectation. $\widehat{P}_{k,h}$ need not itself be a probability kernel.
$\beta^r_{k,e,h}$, $\beta^g_{k,\ell,h}$, $\beta^P_{k,h}$	Parameter-confidence radii. Unindexed radii are their maxima over the analyzed episodes and factors.
$\overline{\beta}(d,n;B,R)$	Deterministic upper radius for dimension d and at most n observations, defined in (45).
$b^r_{k,h}$, $b^P_{k,h}$, $U^g_{k,\ell,h}$	Reward and transition prediction half-widths, and the constraint upper confidence bound.
$\widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s)$, $\widehat{\mathcal{A}}_{k,h}(s)$	Actions certified by local initial or statistical information, and their union with the whole fallback.
$\widetilde{Q}_{k,h}$, $Q_{k,h}$	Unclipped optimistic score and score clipped to $[0, H-h+1]$.

Symbol	Meaning
$\mathcal{X}_h^\star$, χ_k, B_K	Comparator occupancy support, uncertified-episode indicator, and total number of such episodes.
Δ_{safe}, n_g	Margin for initially uncertified comparator tuples, and observation budget per constraint and step.
δ, α, $\mathcal{E}$	Overall failure budget, per-regression budget $\delta/[2H(F+m+1)]$, and simultaneous parameter-confidence event.
$\Delta_{k,h}$, $f_{k,h}$, $Z_{k,h}$	Optimistic-minus-executed value gap, its positive part, and bounded martingale differences in the regret proof.
$\mathfrak{M}_{K,H}(\delta)$, $\varepsilon_{k,h}$	Trajectory concentration and uniform score-optimization error. Neither permits a safety violation.
$\mathcal{F}$, $r_{\max}$, ω	Full planning factors, maximum action-scope size, and induced elimination width.
I_f, J_f, $\mathcal{F}_{k,h}^Q$	Action scope, state-input scope, and factors of the complete unclipped score.
$\mathcal{T}=(\mathcal{B},\mathcal{E}_\mathcal{T})$, B_b, S_b	Junction tree, bag variables, and parent separators.
u_b, $b(f)$, $b(i)$, $A_{\max}$	Bag host, factor's assigned bag, action's output bag, and maximum individual action-set size.
$p(b)$, $\mathrm{ch}(b)$, F_b	Parent bag, child bags, and sum of score and hard factors assigned to a bag.
Q_b^0, $Q_{b\uparrow}^0$	Fallback score from factors assigned to a bag and its subtree.
z_ℓ, $\mathcal{Z}_\ell$, $\mathcal{Z}_{\ell,h}^0$	Local state–action tuple, configuration set, and sound initial local certificates.
M_ℓ, $n_{k,\ell,h}(z)$, $\overline{g}_{k,\ell,h}(z)$	Local table size, entry observation count, and sample mean.
L_ℓ, q_ℓ, $\mathcal{E}_{\mathrm{tab}}$	Tabular confidence logarithm, sufficient certification count, and combined tabular-safety/reward/transition confidence event.
$S_{\max}$, r_g	Maximum local state-alphabet size and maximum safety-scope size in the local-table comparison.
R_h, R_0, r_{in}, D_G, r_{out}	Permitted rounds, fallback rounds, input distance, routed tree height, and output distance.
$\mathcal{D}_s$, $\mathcal{D}_0$, W_s	Distinct state and fallback source–host demands, and routed state-input bits.
L_f, z_f, ζ, m_Q	Uniform absolute score-factor bound, quantized integer factor, quantization step, and number of score factors.
Z_ζ, b_ζ, T_b	Integer score-magnitude bound, bits per score word, and number of separator assignments.
$\mathcal{M}$, b_p, $\mathcal{R}_p$, W_e, c_e	Routed messages, payload bits, physical paths, per-edge load, and directed edge capacity.
$\mathfrak{D}_i(\Pi)$, $\mathrm{Gap}_i(\Pi)$	Feasible unilateral protocol replacements and their improvement gap from ρ.
$\mathcal{X}_{k,h}$, $\mathcal{A}_{k,h}^{\mathrm{rec}}$	Recursively certified states and actions in the optional continuation extension.
$\widehat{\Theta}_{k,h}^P$, $\mathcal{C}_{k,h}^P$	Valid transition parameters inside the ellipsoid and the corresponding nonempty kernel uncertainty set.
η, δ_{conf}, δ_{exit}	Optional continuation escape tolerance, joint confidence-failure bound, and allocated trajectory escape budget.

A.1 Algorithm inputs and explicit parameter choices

The bounds B_r, B_g, B_P, R_r, R_g must satisfy Assumption 3.1. Their validity cannot be inferred from the same finite data merely by substituting empirical norms or variances. State and action sets, features, scopes, the communication graph and alphabet, permitted rounds, the fallback, and

sound initial certificates are also model inputs. The unknown parameters, optimal safe protocol, and comparator margin are not inputs to the algorithm.

Table 2: Explicit numerical parameters and their roles.

Quantity	Definition and use
H, K, T, δ	Given horizon H, episode count K, and failure budget $0 < \delta < 1$. The $T = KH$ environment steps exclude communication.
λ, R_P	$\lambda = 1$ in every ridge regression. $R_P = 1$ is valid because the normalized transition response lies in $[0, 1]$ on every history.
α	$\delta/[2H(F + m + 1)]$, allocating $\delta/2$ to all $HF + Hm + H$ regression processes. Regret concentration uses the remaining $\delta/2$.
$\beta^r_{k,e,h}$, $\beta^g_{k,\ell,h}$, $\beta^P_{k,h}$	Equations (32)–(33). At empty data, $\beta(I_d; B, R) = B + R\sqrt{2\log(1/\alpha)}$. No additional confidence multiplier is tuned.
$b^r_{k,h}$, $b^P_{k,h}$	Prediction half-widths in (11)–(12). The factor H reverses normalization of the transition regression.
$U^g_{k,\ell,h}$	Estimated constraint plus its prediction half-width, as in (13). Its acceptance threshold is exactly zero.
n_g	An observation bound per (ℓ, h) through episode K. If the auxiliary-feedback model allows at most n_{aux} extra observations per pair, take $n_g = K + n_{\mathrm{aux}}$. Trajectory-only feedback has $n_g = K$. Only the tabular variant requires this bound as input.
L_ℓ, q_ℓ	$L_\ell = \log(2M_\ell n_g/\alpha)$ enters the tabular certificate. $q_\ell = \max\{1, \lceil 8R_g^2 L_\ell/\Delta^2_{\mathrm{safe}}\rceil\}$ is a sufficient sample count in the proof. The learner does not use q_ℓ or Δ_{safe}.
Tie rule	Prefer the whole fallback when its score ties the certified optimum. Resolve other ties using the solver's fixed action and bag order, with conditional backtracking. The same rule is used in precomputation and execution.
$\varepsilon_{k,h}$	Zero for the exact planner. For an approximate planner it is a given uniform loss bound for the original, unclipped score. The achieved score defines the backward value.
L_f, $\zeta_{k,h}$	Take L_f to be the maximum absolute value of factor f over its finite local input domain, or any given upper bound. For the optional integer-score planner, $\zeta_{k,h} = 1/[T\max\{1, m_{Q,k,h}\}]$ bounds total optimization loss by one. Hard-feasibility indicators remain exact.
$R_0, r_{\mathrm{in}}, D_G, r_{\mathrm{out}}$	Fallback's round cost and the route quantities in (99)–(100). The sufficient vector-message budget $R_0 + r_{\mathrm{in}} + 2D_G + r_{\mathrm{out}}$ must not exceed the permitted R_h rounds.

The auxiliary-feedback mechanism is part of the observation model, not an implicit oracle invoked on rejected actions. The certification indicators χ_k, their sum B_K, the margin Δ_{safe}, and coverage witnesses appear only in the analysis. Hosts, routes, elimination order, and encodings are computational inputs. Appendices E and I specify their costs and correctness conditions.

A.2 Regression estimators and confidence radii

Predictability means that a feature is fixed before its response is observed. For response noise ε, conditional R-sub-Gaussianity means $\mathbb{E}[\exp(u\varepsilon) \mid \mathcal{H}_{\mathrm{pre}}] \le \exp(u^2R^2/2)$ for every $u \in \mathbb{R}$. The conditioning history $\mathcal{H}_{\mathrm{pre}}$ contains the feature and all previously revealed information. Simultaneous factor noises need not be independent.

For a d-dimensional regression, let $x_t \in \mathbb{R}^d$ be the feature and Y_t the observed response, whose conditional mean is $\langle x_t, \theta\rangle$ for an unknown parameter θ. Let $\mathcal{D}_k$ index the observations available before episode k, with distinct indices for repeated measurements. Fix regularization $\lambda = 1$ and define the design matrix and ridge estimate

$$\Lambda_k = I_d + \sum\nolimits_{t\in\mathcal{D}_k} x_t x_t^\top, \qquad \widehat{\theta}_k = \Lambda_k^{-1} \sum\nolimits_{t\in\mathcal{D}_k} x_t Y_t. \tag{31}$$

Here I_d is the d-dimensional identity matrix. It makes every design invertible, including at initialization, when $\Lambda_1 = I_d$ and $\widehat{\theta}_1 = 0$. For reward factor (e, h), use $x^\tau_{r,e,h} = \phi^r_{e,h}(s^\tau_{e,h}, \mathbf{a}^\tau_{e,h})$ and measured factor reward $Y^\tau_{r,e,h}$ for $\tau < k$. For constraint (ℓ, h), $\mathcal{D}^g_{k,\ell,h}$ indexes all safely obtained observations, including executed-action feedback. Applying (31) defines $(\Lambda^r_{k,e,h}, \widehat{\theta}^r_{k,e,h})$ and $(\Lambda^g_{k,\ell,h}, \widehat{\theta}^g_{k,\ell,h})$.

For transition estimation, apply (31) to the historical-value samples in (10), for $\tau < k$, to obtain $\Lambda^P_{k,h}$ and $\widehat{\theta}^P_{k,h}$.

Allocate confidence failure $\alpha = \delta/[2H(F+m+1)]$ to each regression process and set the normalized transition-noise bound $R_P = 1$. For design Λ, parameter norm bound B, and noise bound R, define the parameter-confidence radius

$$\beta(\Lambda; B, R) = B + R\sqrt{\log\det\Lambda + 2\log(1/\alpha)}. \tag{32}$$

The three radii are explicitly

$$\beta^r_{k,e,h} = \beta(\Lambda^r_{k,e,h}; B_r, R_r), \quad \beta^g_{k,\ell,h} = \beta(\Lambda^g_{k,\ell,h}; B_g, R_g), \quad \beta^P_{k,h} = \beta(\Lambda^P_{k,h}; B_P, R_P). \tag{33}$$

Appendix C.1 proves simultaneous parameter confidence for these radii. All features in each regression are fixed before their corresponding responses are observed.

A.3 Tabular safety confidence bounds

For the variant in Proposition 5.6, observations are labeled local entries with mean $g_{\ell,h}(z)$. Let $M_\ell = |\mathcal{Z}_\ell|$, and let $n_{k,\ell,h}(z)$ and $\overline{g}_{k,\ell,h}(z)$ be the count and sample mean of observations of entry z before episode k. A given integer budget $n_g \geq K$ bounds all observations per (ℓ, h) through episode K. With the failure allocation $\alpha = \delta/[2H(F+m+1)]$, set $L_\ell = \log(2M_\ell n_g/\alpha)$ and define

$$U^{g,\mathrm{tab}}_{k,\ell,h}(z) = \begin{cases} \overline{g}_{k,\ell,h}(z) + R_g\sqrt{2L_\ell/n_{k,\ell,h}(z)}, & n_{k,\ell,h}(z) > 0, \\ +\infty, & n_{k,\ell,h}(z) = 0. \end{cases} \tag{34}$$

This tabular upper confidence bound replaces (13) in the certification test. Known local certificates and the whole fallback are retained. The infinite value prevents an unobserved entry from passing the statistical test. Appendix J.3 proves confidence simultaneously over entries and observation counts. Proposition 5.6 uses the margin-dependent count q_ℓ only in its analysis.

B Related Work and Single-Agent Comparisons

B.1 Single-agent specialization and the meaning of safety

Constraint semantics. Instantaneous and cumulative constraints specify different quantities. For this comparison, c_h denotes a state–action cost, τ_h a prescribed per-step threshold, C a cumulative budget, and π a possibly randomized single-agent policy. An actionwise requirement is

$$\mathbb{P}\big(c_h(s^k_h, a^k_h) \leq \tau_h \text{ for every } k, h\big) \geq 1 - \delta. \tag{35}$$

The probability is over the complete learning interaction. The constraint concerns the true cost, not the noise in its measurement. In contrast, an instantaneous expectation constraint restricts the action distribution through

$$\mathbb{E}_{a\sim\pi_h(\cdot|s)}[c_h(s, a)] \leq \tau_h. \tag{36}$$

A distribution satisfying (36) can assign positive probability to actions whose individual costs exceed τ_h. The expected cumulative cost $J_c(\pi)$ is constrained by the budget C through

$$J_c(\pi) := \mathbb{E}_{\rho,P,\pi}\left[\sum\nolimits_{h=1}^{H} c_h(s_h, a_h)\right] \leq C. \tag{37}$$

Even if (37) holds for every policy used during learning with high probability, it does not impose a per-action threshold. For example, a two-step trajectory with costs 0.8 and 0 satisfies a cumulative budget $C = 0.8$ but violates the instantaneous threshold 0.4 at its first step. Likewise, a bound on cumulative constraint violation may permit violations during learning. A pathwise cumulative budget $\sum_h c_h \leq C$ constrains each trajectory but likewise permits uneven costs across steps. A reward

penalty $V_1^\pi(\rho) - \eta_{\text{pen}} J_c(\pi)$, with penalty weight $\eta_{\text{pen}} \geq 0$, imposes no feasibility requirement by itself. Our guarantees use the actionwise condition (35).

The zero-violation results of Liu et al. [15] and Bura et al. [16] enforce (37) for every selected policy under a known strictly feasible baseline policy. They therefore provide stringent guarantees for cumulative budgets, with a different feasible-policy class from the present actionwise problem. Bi-level provisioning and scheduling [38] couples online budget decisions to a lower-level CMDP through expected episode consumption. Its high-probability feasibility guarantee concerns those episode budgets. The present requirement constrains each executed joint action. Similarly, the randomized-policy extension in Amani et al. [4, Section 4] uses (36), whereas its deterministic-policy algorithm uses actionwise constraints. Randomization over safe actions preserves actionwise safety.

Hard constraints in online optimization. Hard feasibility also arises outside RL. Shi et al. [18] combine known demand constraints with hard ramp limits and a specified uncertainty set for future inputs. Shi et al. [19] study revealed covering constraints and look-ahead under switching costs. Both compare the online cost with an offline dynamic decision sequence through a competitive ratio. In our model, the constraint parameters and Markov dynamics are unknown, and safely obtained observations determine which actions can be certified. The comparator is an optimal admissible safe policy. These information assumptions and benchmarks differ, precluding a direct rate comparison with Theorem 5.2.

Single-agent specialization. For $N = 1$, the joint action becomes an ordinary action and the communication protocol becomes a policy. Taking one reward factor, $F = 1$ and $D_r = d_r$, Corollary 5.4 specializes to

$$\mathrm{Reg}(K) \leq \widetilde{O}\Big(\beta^r \sqrt{H d_r T} + H\beta^P \sqrt{H d_P T} + H^2 m d_g (\beta^g)^2 / \Delta_{\text{safe}}^2 + H\sqrt{T}\Big), \qquad T = KH. \tag{38}$$

Setting $N = 1$ does not imply $m = 1$ or $F = 1$. We choose one reward factor and retain all m constraints. The transition radius β^P corresponds to the normalized regression in (10). The specialization retains the known safe fallback at every modeled state and the comparator margin and feedback-coverage condition of Assumption 5.3. The zero-violation theorem does not require the latter condition, but the sublinear-regret corollary does.

Comparison with instantaneous-safe MDP results. The parameterizations and sources of safe information differ across the closest single-agent results. In a linear MDP, $P_h(s' \mid s, a) = \langle \phi(s,a), \mu_h(s') \rangle$ has a known state–action feature and unknown vector-valued measure. In the linear mixture model used here, $P_h(s' \mid s, a) = \langle \phi_h^P(s' \mid s, a), \theta_h^P \rangle$ has known transition features and a finite-dimensional unknown parameter. Dimensions from these two representations cannot be identified without an additional representation argument.

Table 3: Assumptions relevant to the single-agent comparison. The instantaneous-safety rows concern the latent constraint value, with noisy measurements used for learning. The symbols τ and $\bar{c}$ denote the cited works' safety thresholds.

Work	Safety and dynamics	Observations and initial safety	Regret conditions
Amani et al. [4], deterministic algorithm	Actionwise $c_h(s,a) \leq \tau$ in a linear MDP.	Unknown rewards, costs, and transitions. Executed actions give exact reward and noisy cost feedback. A known action at every state has known cost strictly below τ.	Statewise feature sets are star-convex about the safe feature. A covering-number correction is given by Roknilamouki et al. [6].
Shi et al. [5]	$c_h(s,a,s') \leq \bar{c}$ for realized transitions. Unsafe states are permitted in a linear mixture model.	Known rewards, unknown dynamics and transition costs, and noisy cost feedback. Exact transition supports and a seed safe subgraph are known.	Star-convex subgraph features, Lipschitz conditions on rewards and safe transitions, and a positive safety reserve after accounting for successor-feature variation.

Work	Safety and dynamics	Observations and initial safety	Regret conditions
Roknilamouki et al. [6]	Actionwise $c_h(s,a) \leq \tau$ in a linear MDP with continuous actions.	Unknown rewards, costs, and transitions. Noisy reward and cost feedback. A zero-cost action is known at each state.	Star-convexity, or a Local Point Assumption providing a safe neighborhood and suitable segments near the constraint boundary. The latter case includes an initial safe exploration phase.
Present $N = 1$ specialization	All $g_{\ell,h}(s,a) \leq 0$ on executed actions. Finite states/actions with linear mixture kernels.	Unknown rewards, constraints, and transitions. Noisy reward and constraint feedback. A safe fallback is available at every state. Any auxiliary safety observations must be specified and obtained safely.	Exact planning, strict margins on initially uncertified tuples of nonfallback comparator actions, and explicit growth of the observed constraint designs in uncertified comparator directions. No interpolation is required.

The geometric conditions in the prior actionwise results support safe exploration toward an optimal action, including an optimum on the constraint boundary. Our corollary instead requires a strictly positive margin for each initially uncertified tuple of a nonfallback comparator action and a separate condition on information obtained safely. Known local certificates can already cover boundary tuples, as the example in Appendix J.2 demonstrates. A margin alone does not provide information in an unobserved feature direction. The finite-action model removes the need for physically available actions along a continuous line segment, but explicitly requires informative safe observations. These assumptions therefore do not uniformly weaken the single-agent assumptions. Any acquisition cost for auxiliary safety observations is outside the environment-episode regret and must be included when comparing total learning resources.

The cited instantaneous-safe algorithms obtain regret of square-root order in the number of episodes under their respective assumptions, up to logarithmic factors and their stated exploration costs. In particular, Roknilamouki et al. [6] give a corrected analysis for the star-convex setting of Amani et al. [4] and extend the analysis to a specified non-star-convex class. Equation (38) provides the same qualitative combination of sublinear regret and high-probability actionwise safety under the stated margin and coverage assumptions. The comparison is qualitative and establishes neither a sharper single-agent rate nor a guarantee covering all of these models.

B.2 Multi-agent coordination and solution concepts

Local observations and latent-state games. The distinction between cooperative, competitive, and general-sum objectives is standard in MARL [2]. Local observations also require a precise information model. Liu et al. [39] study weakly revealing partially observable Markov games, using optimistic maximum-likelihood estimation to learn equilibria or compete with adversarial opponents. Here, centralized training observes the joint state. Each agent observes its own state component at execution and communicates before the joint action. The resulting protocol is a Markov map of the joint state, even though no agent need know that state in full. Our Bellman recursion relies on this model and the exact implementability condition. It does not establish a result for general latent-state games or arbitrary history-dependent decentralized policies.

Structured coordination and equilibrium benchmarks. Factored planning and coordinated reinforcement learning already retain multi-agent structure within a centralized MDP representation and use message passing to select a joint action from local value factors [7, 8]. Our use of variable elimination and induced-width complexity builds on this literature. For decentralized safety enforcement, Melcer et al. [22] decompose a given centralized shield using a sound safety-relevant environment abstraction. Here, local constraint parameters are learned from specified feedback, and initial and learned certificates enter the same local feasibility factors. These factors support team-regret analysis with safety during learning and execution through admissible communication.

Constrained equilibrium formulations encompass several safety notions. Alatur et al. [28] learn Nash policies in Markov potential games with coupled expected cumulative-cost constraints. McMahan [31]

imposes almost-sure bounds on every prefix of cumulative cost and reduces equilibrium computation to action-constrained Markov games. The computational guarantees take the game model as input, and the paper also discusses model-based learning extensions. Statewise constraints occur in equilibrium-based safe MARL as well. Li and Azizan [32] study controlled invariant sets and convergence to a generalized Nash equilibrium in statewise-constrained cooperative Markov games. A generalized Nash equilibrium is defined through feasible unilateral deviations. A globally optimal admissible safe team protocol maximizes the common return over the full admissible class. The two benchmarks need not coincide. Corollary 5.7 bounds the average feasible unilateral-deviation gap by team regret for the common payoff. The converse counterexample in Appendix J shows that unilateral stability need not imply team optimality. This consequence concerns admissible safe protocol replacements from the initial distribution. Convergence of independent player updates and Markov-perfect equilibrium conditions at every state require separate analyses.

Hierarchical information and Stackelberg learning. Kao et al. [33] study cooperative bandits and MDPs in which a follower observes the leader's action before choosing its own. Zhong et al. [34] study Stackelberg–Nash equilibria with myopically rational followers, and Zhao et al. [35] analyze repeated cooperative Stackelberg games with an omniscient follower. These information and response structures differ from an elimination order in a cooperative planner. Exact elimination computes a joint maximizing assignment, after which all agents execute simultaneously. No agent optimizes against a strategic follower response. Exact elimination orders preserve the feasible set and optimal value but can change computation, communication, and the selected tied maximizer.

Differential games and reach-avoid control. Fridovich-Keil et al. [25] develop iterative linear-quadratic approximations for solving nonlinear general-sum differential games. This approach models strategic interaction through repeated local game approximations. Chen et al. [26] combine pairwise reach-avoid solutions with maximum matching to obtain defending-team performance guarantees in multiplayer differential games. These methods concern game solving and reachability for specified dynamical models. Our analysis concerns statistical learning with unknown reward, transition, and constraint parameters and accounts for safety during data collection.

C Proofs of the Safety and Regret Guarantees

We use the action sets $\widehat{\mathcal{A}}_{k,h}(s) = \widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s) \cup \{\mathbf{a}^0_h(s)\}$, where $\mathbf{a}^0_h(s)$ is the known-safe fallback action. All value functions are defined on the full state space. The terminal value is zero, and the optimistic backup is clipped to $[0, H-h+1]$. Exact optimization and a fixed tie-breaking rule determine the executed admissible protocol Π^k before episode k. Certificate soundness guarantees safety independently of comparator availability. The latter affects regret through the episodes in which the certified action sets exclude a comparator action.

C.1 Confidence Event and Preliminary Lemmas

For each reward factor, let

$$x^\tau_{r,e,h} = \phi^r_{e,h}(s^\tau_{e,h}, \mathbf{a}^\tau_{e,h}), \qquad Y^\tau_{r,e,h} = \langle x^\tau_{r,e,h}, \theta^r_{e,h}\rangle + \varepsilon^\tau_{r,e,h}. \tag{39}$$

Here $x^\tau_{r,e,h}$ is the observed local feature, $Y^\tau_{r,e,h}$ is its factor reward, and $\varepsilon^\tau_{r,e,h}$ is the centered measurement noise. The identity-regularized design matrix and least-squares estimate are

$$\Lambda^r_{k,e,h} = I + \sum\nolimits_{\tau<k} x^\tau_{r,e,h}(x^\tau_{r,e,h})^\top, \qquad \widehat{\theta}^r_{k,e,h} = (\Lambda^r_{k,e,h})^{-1} \sum\nolimits_{\tau<k} x^\tau_{r,e,h} Y^\tau_{r,e,h}. \tag{40}$$

For constraint (ℓ, h), $\mathcal{D}^g_{k,\ell,h}$ indexes observations available before episode k, including auxiliary measurements permitted by the feedback model. Repeated measurements have distinct indices. Each observation has a predictable feature x_t with $\|x_t\|_2 \le 1$ and response $Y_t = \langle x_t, \theta^g_{\ell,h}\rangle + \varepsilon_t$. The design matrix and ridge estimate are

$$\Lambda^g_{k,\ell,h} = I + \sum_{t\in\mathcal{D}^g_{k,\ell,h}} x_t x_t^\top, \qquad \widehat{\theta}^g_{k,\ell,h} = (\Lambda^g_{k,\ell,h})^{-1} \sum_{t\in\mathcal{D}^g_{k,\ell,h}} x_t Y_t. \tag{41}$$

The noise sequences are conditionally R_r- and R_g-sub-Gaussian, respectively. Their parameters satisfy $\|\theta^r_{e,h}\|_2 \le B_r$ and $\|\theta^g_{\ell,h}\|_2 \le B_g$.

For transitions, the normalized feature and response retain the value function fixed before that transition,

$$x^\tau_{P,h} = \psi_h^{V_{\tau,h+1}}(s^\tau_h, \mathbf{a}^\tau_h)/H, \qquad Y^\tau_{P,h} = V_{\tau,h+1}(s^\tau_{h+1})/H. \tag{42}$$

The design and estimator are

$$\Lambda^P_{k,h} = I + \sum\nolimits_{\tau<k} x^\tau_{P,h}(x^\tau_{P,h})^\top, \qquad \widehat{\theta}^P_{k,h} = (\Lambda^P_{k,h})^{-1} \sum\nolimits_{\tau<k} x^\tau_{P,h} Y^\tau_{P,h}. \tag{43}$$

Both $V_{\tau,h+1}$ and $x^\tau_{P,h}$ are measurable before s^τ_{h+1} is sampled. Let $\mathcal{H}^{\mathrm{pre}}_{\tau,h}$ denote the history immediately before this sample, including the current state and selected action. By linear-mixture realizability,

$$\mathbb{E}[Y^\tau_{P,h} \mid \mathcal{H}^{\mathrm{pre}}_{\tau,h}] = \langle x^\tau_{P,h}, \theta^P_h \rangle.$$

Since the value function is clipped on every history, including histories outside the confidence event, $Y^\tau_{P,h} \in [0,1]$ and $\|x^\tau_{P,h}\|_2 \le 1$. The centered response is conditionally R_P-sub-Gaussian for the declared conservative choice $R_P := 1$. The bound $\|\theta^P_h\|_2 \le B_P$ is part of Assumption 3.1.

Allocate failure probability $\alpha = \delta/[2H(F+m+1)]$ to each regression process. For a process with design Λ, parameter bound B, and noise parameter R, use

$$\beta(\Lambda; B, R) = B + R\sqrt{\log\det(\Lambda) + 2\log(1/\alpha)}. \tag{44}$$

The radii $\beta^r_{k,e,h}$, $\beta^g_{k,\ell,h}$, and $\beta^P_{k,h}$ use (B_r, R_r), (B_g, R_g), and (B_P, R_P), respectively. Any larger valid radii also satisfy the results below.

For explicit bounds that do not depend on the realized design, define

$$\overline{\beta}(d, n; B, R) := B + R\sqrt{d\log(1 + n/d) + 2\log(1/\alpha)}, \qquad d \ge 1, \quad n \ge 0. \tag{45}$$

Here d is the regression dimension and n bounds its sample count. For $\Lambda = I + \sum_{t=1}^{n'} x_t x_t^\top$, with $n' \le n$ and $\|x_t\|_2 \le 1$, the eigenvalue arithmetic–geometric mean inequality gives $\log\det\Lambda \le d\log(1 + n/d)$. Consequently, the maxima over the first K episodes used in Theorem 5.2 satisfy

$$\begin{aligned} \beta^r &\le \max_{e\in\mathcal{E}_r} \overline{\beta}(d_{r,e}, K; B_r, R_r), \\ \beta^P &\le \overline{\beta}(d_P, K; B_P, R_P), \qquad \beta^g \le \overline{\beta}(d_g, n_g; B_g, R_g). \end{aligned} \tag{46}$$

Here n_g bounds the observations in any one constraint regression. The algorithm uses the design-dependent radii in (32). These envelopes make their finite-sample and asymptotic dependence explicit. The comparator margin Δ_{safe} and failure count B_K are used only in the analysis.

Define the local predictions $\widehat{r}_{k,e,h} = \langle \phi^r_{e,h}, \widehat{\theta}^r_{k,e,h} \rangle$ and $\widehat{g}_{k,\ell,h} = \langle \phi^g_{\ell,h}, \widehat{\theta}^g_{k,\ell,h} \rangle$ with the same local arguments as the true functions.

Lemma C.1 (Simultaneous parameter confidence)**.** *With probability at least $1 - \delta/2$, the event*

$$\begin{aligned} \mathcal{E} := \big\{ &\|\widehat{\theta}^r_{k,e,h} - \theta^r_{e,h}\|_{\Lambda^r_{k,e,h}} \le \beta^r_{k,e,h}, \quad \|\widehat{\theta}^g_{k,\ell,h} - \theta^g_{\ell,h}\|_{\Lambda^g_{k,\ell,h}} \le \beta^g_{k,\ell,h}, \\ &\|\widehat{\theta}^P_{k,h} - \theta^P_h\|_{\Lambda^P_{k,h}} \le \beta^P_{k,h} \textit{ for all } k, h, e, \ell \big\} \end{aligned} \tag{47}$$

holds, with indices ranging over $k \in [K]$, $h \in [H]$, $e \in \mathcal{E}_r$, and $\ell \in [m]$. Consequently, on $\mathcal{E}$, for every state, joint action, and bounded value function V,

$$\begin{aligned} |r_{e,h} - \widehat{r}_{k,e,h}| &\le \beta^r_{k,e,h} \|\phi^r_{e,h}\|_{(\Lambda^r_{k,e,h})^{-1}}, \\ |g_{\ell,h} - \widehat{g}_{k,\ell,h}| &\le \beta^g_{k,\ell,h} \|\phi^g_{\ell,h}\|_{(\Lambda^g_{k,\ell,h})^{-1}}, \\ |P_h V - \widehat{P}_{k,h} V| &\le H\beta^P_{k,h} \|\psi^V_h / H\|_{(\Lambda^P_{k,h})^{-1}}. \end{aligned}$$

Arguments of the functions have been suppressed in these three inequalities.

Proof of Lemma C.1. For a predictable linear-regression sequence, write $\varepsilon_t = Y_t - \langle x_t, \theta \rangle$ and define the noise-feature sum $S = \sum_t x_t \varepsilon_t$. With regularization I, the ridge error obeys

$$\widehat{\theta} - \theta = \Lambda^{-1}(S - \theta), \qquad \|\widehat{\theta} - \theta\|_\Lambda \le \|S\|_{\Lambda^{-1}} + \|\theta\|_{\Lambda^{-1}} \le \|S\|_{\Lambda^{-1}} + B.$$

The last step uses $\Lambda \succeq I$ and $\|\theta\|_2 \le B$. For conditionally R-sub-Gaussian noise, the self-normalized inequality [36, Theorems 1 and 2] therefore gives, simultaneously at every sample size,

$$\|\widehat{\theta} - \theta\|_\Lambda \le B + R\sqrt{\log\det\Lambda + 2\log(1/\alpha)}$$

with probability at least $1-\alpha$. Apply this result to the HF reward processes, the Hm constraint processes, and the H transition processes. The preceding predictability and boundedness argument verifies the transition-regression hypotheses. The union bound over these processes has failure probability at most $H(F+m+1)\alpha = \delta/2$.

For any feature vector x, Cauchy–Schwarz gives

$$|\langle x, \widehat{\theta} - \theta\rangle| \le \|\widehat{\theta} - \theta\|_\Lambda \|x\|_{\Lambda^{-1}}.$$

Apply this inequality to each reward feature, constraint feature, and $x = \psi_h^V / H$. The parameter event implies the prediction bounds simultaneously for every such vector. In particular, no union bound over value functions or states is required. The same statement applies to a value function selected from the data after the parameter event is established. □

At the fixed next-step value, write $b^P_{k,h}(s,\mathbf{a}) = b^P_{k,h}(s,\mathbf{a}; V_{k,h+1})$. The reward and transition confidence half-widths are

$$b^r_{k,h}(s,\mathbf{a}) := \sum\nolimits_{e\in\mathcal{E}_r} \beta^r_{k,e,h} \|\phi^r_{e,h}(s_e,\mathbf{a}_e)\|_{(\Lambda^r_{k,e,h})^{-1}}, \tag{48}$$

$$b^P_{k,h}(s,\mathbf{a}) := H\beta^P_{k,h} \|\psi_h^{V_{k,h+1}}(s,\mathbf{a})/H\|_{(\Lambda^P_{k,h})^{-1}}. \tag{49}$$

Thus, on $\mathcal{E}$, the optimistic reward and transition-value estimates overestimate their true counterparts by amounts in $[0, 2b^r_{k,h}]$ and $[0, 2b^P_{k,h}]$, respectively.

C.2 Proof of Theorem 5.1 (Zero Violation)

Lemma C.2 (Soundness of the coupled certificates)**.** *On $\mathcal{E}$, every action in $\widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s)$ satisfies $g_{\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell}) \le 0$ for every ℓ. Every action in $\widehat{\mathcal{A}}_{k,h}(s)$ is therefore safe.*

Proof of Lemma C.2. Lemma C.1 implies $g_{\ell,h} \le U^g_{k,\ell,h}$ for every constraint. Membership in the immediate certified set provides either $z_\ell \in \mathcal{Z}^0_{\ell,h}$, which is safe by the known certificate, or $U^g_{k,\ell,h} \le 0$, which implies safety on $\mathcal{E}$. The remaining candidate, $\mathbf{a}^0_h(s)$, is safe by the initial safety certificate. □

Proof of Theorem 5.1. The candidate set is nonempty at every state because it contains the fallback action. The implemented protocol selects a joint action from this set. Lemma C.2 therefore establishes safety at every episode and step on $\mathcal{E}$, including states reached after previous exploratory actions. Lemma C.1 bounds the failure probability by $\delta/2 \le \delta$. □

Lemma C.3 (Optimism for an available comparator)**.** *On $\mathcal{E}$, if an admissible protocol Π selects candidate actions at every state-action pair in its reachable support, then $V_{k,h}(s) \ge V^\Pi_h(s)$ at every state reachable under Π at step h. Moreover, $V_{k,h}(s) \ge V^{\Pi^k}_h(s)$ for every h, s.*

Proof of Lemma C.3. The terminal values are zero. Suppose the assertion for Π holds at step $h+1$, and let $(s,\mathbf{a})$ be in its step-h support. Every successor with positive transition probability is reachable under Π. Monotonicity of the true transition kernel and confidence give

$$r_h(s,\mathbf{a}) + P_h V^\Pi_{h+1}(s,\mathbf{a}) \le r_h(s,\mathbf{a}) + P_h V_{k,h+1}(s,\mathbf{a}) \le \widetilde{Q}_{k,h}(s,\mathbf{a}).$$

The left-hand side belongs to $[0, H-h+1]$, so clipping the right-hand side to this interval preserves the inequality. Exact maximization over the candidate actions gives $V_{k,h}(s) \ge Q_{k,h}(s,\mathbf{a})$. Because the protocol is deterministic, the resulting inequality is the required comparison with its Bellman value. For Π^k, its selected actions are candidates at every state and its value recursion is defined on the full state space, so the same induction applies at every s. □

C.3 Proof of Theorem 5.2 (Regret Bound)

Let $b^r_{k,h} = b^r_{k,h}(s^k_h, \mathbf{a}^k_h)$ and define $b^P_{k,h}$ similarly. Recall χ_k from (22). Sums over k, h below range over $[K] \times [H]$ unless specified otherwise. Because $0 \le V_{k,1}(\rho) \le H$, Lemma C.3 implies, on $\mathcal{E}$,

$$V_1^{\Pi^\star}(\rho) - V_1^{\Pi^k}(\rho) \le V_{k,1}(\rho) - V_1^{\Pi^k}(\rho) + H\chi_k. \tag{50}$$

The indicator χ_k concerns the comparator's complete reachable support, including states that the learner never visits.

Lemma C.4 (High-probability regret decomposition)**.** *With probability at least $1-\delta$, jointly with $\mathcal{E}$,*

$$\mathrm{Reg}(K) \le 2\sum\nolimits_{k,h} b^r_{k,h} + 2\sum\nolimits_{k,h} b^P_{k,h} + HB_K + \mathfrak{M}_{K,H}(\delta), \tag{51}$$

$$\mathfrak{M}_{K,H}(\delta) := \sqrt{KH(H+1)(2H+1)\log(2/\delta)/3} \le H\sqrt{2T\log(2/\delta)}. \tag{52}$$

Here $\mathfrak{M}_{K,H}(\delta)$ bounds sampling error from the initial states and transitions.

Proof of Lemma C.4. Define the optimistic value error

$$\Delta_{k,h}(s) := V_{k,h}(s) - V_h^{\Pi^k}(s). \tag{53}$$

On $\mathcal{E}$, Lemma C.3 gives $0 \le \Delta_{k,h}(s) \le H - h + 1$. For the action selected by Π^k, the clipped optimistic Bellman backup and the confidence bounds imply

$$\Delta_{k,h}(s^k_h) \le 2b^r_{k,h} + 2b^P_{k,h} + P_h\Delta_{k,h+1}(s^k_h, \mathbf{a}^k_h). \tag{54}$$

Indeed, the unclipped optimistic backup is at least $r_h + P_hV_{k,h+1} \ge 0$ on $\mathcal{E}$, so its lower clipping is inactive. Upper clipping can only reduce the estimation error. More explicitly, the selected action has $V_{k,h} = Q_{k,h} \le \widetilde{Q}_{k,h}$ and

$$\begin{aligned}\widetilde{Q}_{k,h} - r_h - P_hV_{k,h+1} &= (\widehat{r}_{k,h} + b^r_{k,h} - r_h) \\ &\quad + (\widehat{P}_{k,h}V_{k,h+1} + b^P_{k,h} - P_hV_{k,h+1}) \\ &\le 2b^r_{k,h} + 2b^P_{k,h}.\end{aligned}$$

Subtracting $V_h^{\Pi^k} = r_h + P_hV_{h+1}^{\Pi^k}$ gives (54). All terms use the selected state–action pair.

To apply concentration without conditioning on $\mathcal{E}$, define the nonnegative error function $f_{k,h}(s) = [\Delta_{k,h}(s)]_+$, where $[u]_+ = \max\{u, 0\}$ denotes the positive part. The function is measurable at the start of episode k and satisfies $0 \le f_{k,h}(s) \le H - h + 1$ on every history, because both value functions lie in this interval. On $\mathcal{E}$, $f_{k,h} = \Delta_{k,h}$. Write $f_{k,1}(\rho) = \sum_{s\in\mathcal{S}} \rho(s) f_{k,1}(s)$ and define the initial-state and transition sampling errors

$$\begin{aligned}Z_{k,0} &:= f_{k,1}(\rho) - f_{k,1}(s^k_1), \\ Z_{k,h} &:= P_hf_{k,h+1}(s^k_h, \mathbf{a}^k_h) - f_{k,h+1}(s^k_{h+1}), \qquad h = 1, \dots, H-1.\end{aligned} \tag{55}$$

Conditional on the preceding history, the initial state is drawn from ρ, and each subsequent state is drawn from the true transition kernel. Thus, these variables in temporal order are martingale differences. They satisfy $|Z_{k,0}| \le H$ and $|Z_{k,h}| \le H - h$ unconditionally. Azuma–Hoeffding therefore gives

$$\sum\nolimits_{k=1}^{K}\sum\nolimits_{h=0}^{H-1} Z_{k,h} \le \mathfrak{M}_{K,H}(\delta)$$

with probability at least $1 - \delta/2$. The sum of squared increment bounds is $K\sum_{j=1}^{H} j^2 = KH(H+1)(2H+1)/6$, which gives the closed form in (52).

On $\mathcal{E}$, replacing each conditional next error by its sampled value plus $Z_{k,h}$ in (54) makes the value errors telescope. Since $\Delta_{k,H+1} = 0$,

$$\Delta_{k,1}(\rho) \le 2\sum\nolimits_{h=1}^{H}(b^r_{k,h} + b^P_{k,h}) + \sum\nolimits_{h=0}^{H-1} Z_{k,h}. \tag{56}$$

Summing this inequality and intersecting with the martingale event bounds its last term by $\mathfrak{M}_{K,H}(\delta)$. Combining this with (50) and taking a union bound over the confidence and martingale events proves the claim. The bound $\sum_{j=1}^{H} j^2 \le H^3$ gives the final inequality in (52). □

Lemma C.5 (Cumulative reward and transition widths)**.** *For the actual trajectories,*

$$2\sum\nolimits_{k,h} b^r_{k,h} \le 2\beta^r \sqrt{2FHD_rT\log(1+K)}, \tag{57}$$

$$2\sum\nolimits_{k,h} b^P_{k,h} \le 2H\beta^P \sqrt{2Hd_PT\log(1+K/d_P)}. \tag{58}$$

Proof of Lemma C.5. For vectors $x_1, \dots, x_K \in \mathbb{R}^d$ with $\|x_k\|_2 \le 1$, let $\Lambda_k = I + \sum_{\tau<k} x_\tau x_\tau^\top$ and $u_k = x_k^\top \Lambda_k^{-1} x_k$, the squared confidence width of x_k. Then $0 \le u_k \le 1$. The determinant lemma and $u \le 2\log(1+u)$ for $u \in [0,1]$ imply

$$\sum\nolimits_{k=1}^{K} u_k \le 2\log\det\Lambda_{K+1} \le 2d\log(1+K/d).$$

The last inequality follows from $\mathrm{tr}(\Lambda_{K+1}) \le d+K$ and the arithmetic–geometric mean inequality for its eigenvalues. By Cauchy–Schwarz,

$$\sum\nolimits_{k=1}^{K} \|x_k\|_{\Lambda_k^{-1}} \le \sqrt{2Kd\log(1+K/d)}.$$

Apply this separately at every (e,h) to the reward features. Summing over H steps, using $\log(1+K/d_{r,e}) \le \log(1+K)$, and applying $\sum_e \sqrt{d_{r,e}} \le \sqrt{FD_r}$ gives

$$\sum\nolimits_{k,h,e} \|\phi^r_{e,h}(s^k_{e,h}, \mathbf{a}^k_{e,h})\|_{(\Lambda^r_{k,e,h})^{-1}} \le H\sqrt{2KFD_r\log(1+K)}.$$

Multiplying by $2\beta^r$ and substituting $T = KH$ proves (57).

For transitions, the vectors used in the current width, $\psi_h^{V_{k,h+1}}(s^k_h, \mathbf{a}^k_h)/H$, are exactly the predictable features inserted into $\Lambda^P_{k+1,h}$ after the transition. They have norm at most one. Applying the same bound for each h gives

$$\sum\nolimits_{k,h} \|\psi_h^{V_{k,h+1}}(s^k_h, \mathbf{a}^k_h)/H\|_{(\Lambda^P_{k,h})^{-1}} \le H\sqrt{2Kd_P\log(1+K/d_P)}.$$

Multiply by $2H\beta^P$ and use $T = KH$ to obtain (58). Separate step-indexed matrices ensure that each estimate uses samples of its own transition parameter. □

Proof of Theorem 5.2. Combine Lemmas C.4 and C.5. The same confidence event proves zero violation, so the probability statement is joint with Theorem 5.1. □

C.4 Proof of Corollary 5.4 (Certification Coverage)

A sufficient feedback model. For example, suppose a safe sensing mechanism provides one noisy observation in each coordinate direction of $\theta^g_{\ell,h}$ after every episode, for every (ℓ,h). These observations have features equal to the coordinate vectors and need not execute the candidate actions. Their design increment is I, which dominates $xx^\top$ for every $\|x\|_2 \le 1$. The coverage condition then holds with at most $(d_g+1)K$ constraint observations per pair, including the executed-action measurements. This additional sensing model must be specified separately from trajectory feedback. Its measurement cost remains separate from the stated regret.

Lemma C.6 (Certification of an interior action)**.** *On $\mathcal{E}$, if $g_{\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell}) \le -\Delta_{\mathrm{safe}}$ and*

$$\beta^g_{k,\ell,h}\|\phi^g_{\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell})\|_{(\Lambda^g_{k,\ell,h})^{-1}} \le \Delta_{\mathrm{safe}}/2,$$

then $U^g_{k,\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell}) \le 0$.

Proof of Lemma C.6. Parameter confidence implies

$$U^g_{k,\ell,h} \le g_{\ell,h} + 2\beta^g_{k,\ell,h}\|\phi^g_{\ell,h}\|_{(\Lambda^g_{k,\ell,h})^{-1}} \le -\Delta_{\mathrm{safe}} + \Delta_{\mathrm{safe}} = 0.$$

□

Lemma C.7 (Number of uncertified comparator episodes)**.** *Under Assumption 5.3, on $\mathcal{E}$,*

$$B_K \le \frac{8Hmd_g(\beta^g)^2}{\Delta^2_{\mathrm{safe}}}\log(1+n_g/d_g).$$

Proof of Lemma C.7. If $\beta^g = 0$, confidence gives $U^g_{k,\ell,h} = g_{\ell,h}$ on $\mathcal{E}$ for every tuple and index. Every safe comparator action is then certified, so $B_K = 0$ and the bound holds. Hence suppose $\beta^g > 0$. Assign each episode with $\chi_k = 1$ to one witness (ℓ, h) specified by Assumption 5.3. Denote its feature by x_k and write $u_k = x_k^\top (\Lambda^g_{k,\ell,h})^{-1} x_k$. The witness belongs to a nonfallback comparator action and its tuple lies outside $\mathcal{Z}^0_{\ell,h}$, so its constraint has the stipulated margin. Since its upper confidence bound is positive, Lemma C.6 implies

$$u_k > \Delta^2_{\mathrm{safe}}/[4(\beta^g)^2]. \tag{59}$$

Also $u_k \le 1$, because $\|x_k\|_2 \le 1$ and $\Lambda^g_{k,\ell,h} \succeq I$.

Fix a pair (ℓ, h), and let $\mathcal{I}_{\ell,h}$ be its assigned episodes. The covariance-growth condition and monotonicity of the logarithmic determinant give, for each assigned episode,

$$\log\det \Lambda^g_{k+1,\ell,h} - \log\det \Lambda^g_{k,\ell,h} \ge \log(1 + u_k) \ge u_k/2.$$

All design increments are positive semidefinite, including increments in unassigned episodes. Summing only the assigned increments is therefore at most the full determinant increase. Padding features with zeros when needed gives dimension d_g, and at most n_g observations of norm at most one yield

$$\sum\nolimits_{k \in \mathcal{I}_{\ell,h}} u_k \le 2 \log\det \Lambda^g_{K+1,\ell,h} \le 2 d_g \log(1 + n_g/d_g).$$

Combining this with (59) bounds $|\mathcal{I}_{\ell,h}|$ by $8 d_g (\beta^g)^2 \Delta^{-2}_{\mathrm{safe}} \log(1 + n_g/d_g)$. Each episode with an unavailable comparator action is assigned once, so summing over the Hm pairs proves the result. This argument charges only design growth from observations actually collected. □

Proof of Corollary 5.4. Substitute Lemma C.7 into Theorem 5.2. Fix H, F, m, the regression dimensions, the parameter and noise bounds, $\Delta_{\mathrm{safe}} > 0$, and $\delta \in (0, 1)$. If n_g is polynomial in K, the explicit envelopes in (46) give $\beta^r, \beta^P, \beta^g = O(\sqrt{\log(1 + K)})$. The certification term is therefore $O(\log^2(1 + K))$, while estimation and sampling contribute at most $O(\sqrt{K}\log(1 + K))$. Both contributions are sublinear in K. □

Necessity of informative safe feedback. A globally safe fallback and a positive margin do not suffice for sublinear regret under arbitrary trajectory-only feedback. The following construction makes this information requirement explicit.

Proposition C.8 (Safe feedback need not identify a better safe action)**.** *There is a pair of one-state, one-step linear-constraint problems with a known safe fallback such that every algorithm guaranteeing zero violation with probability at least $1 - \delta$ on both problems has expected regret at least $(1 - \delta)K$ on the problem whose optimal action is strictly safe.*

Proof of Proposition C.8. Let the actions be a_0 and a_c, with known rewards zero and one. Use constraint features $\phi^g(a_0) = (1, 0)^\top$, $\phi^g(a_c) = (0, 1)^\top$, and parameters $\theta^g_\pm = (-1, \pm\gamma)^\top$ for $0 < \gamma \le 1$. Take noiseless constraint observations, so $R_g = 0$, and the valid norm bound $B_g = \sqrt{2}$. Thus a_0 is a known safe fallback in both problems, whereas a_c is safe with margin γ only in the minus problem. Constraint measurements are obtained only by executing the corresponding action. Fallback measurements have identical laws in the two instances.

Write $\mathbb{P}_\pm$ and $\mathbb{E}_\pm$ for probability and expectation under the respective parameters. Let τ be the first episode in which a_c is executed. All observations before τ have identical laws, so $\mathbb{P}_-(\tau \le K) = \mathbb{P}_+(\tau \le K)$. Uniform safety in the plus problem requires the latter probability to be at most δ. If $N_c(K)$ is the number of candidate executions, then

$$\mathbb{E}_- \operatorname{Reg}(K) = K - \mathbb{E}_- N_c(K) \ge K\big(1 - \mathbb{P}_-(\tau \le K)\big) \ge (1 - \delta)K.$$

The features have norm one and the parameter norm is at most $\sqrt{2}$. Thus a positive comparator margin and global fallback do not make the untried candidate identifiable from the available observations. □

Coverage must also concern the comparator's reachable states, which can differ from those visited by the learner. Consider a two-step problem with two known-safe actions at the initial state. Action L gives reward $1/2$ and leads to a state with only a zero-reward fallback. Action R gives reward zero

and leads to a different state with a zero-reward fallback and a candidate action of reward one. The candidate is truly safe with positive margin, but its constraint feature is not observed or spanned by the observations obtained along the L branch. Even with known rewards and transitions, pessimistic planning can select L in every episode while the optimum selects R and the candidate. Regret is then $K/2$. No certification failure occurs at the states actually visited by the learner, although the comparator is uncertified on its own reachable set in every episode. Thus B_K must account for uncertified comparator actions beyond the learner's visited states.

C.5 Regret under Certified Approximate Planning

For a state-uniform error bound $\varepsilon_{k,h} \geq 0$, suppose an implementable planner returns a certified action $\widetilde{\mathbf{a}}_{k,h}(s) \in \widehat{\mathcal{A}}_{k,h}(s)$ satisfying

$$\max_{\mathbf{a} \in \widehat{\mathcal{A}}_{k,h}(s)} \widetilde{Q}_{k,h}(s, \mathbf{a}) - \widetilde{Q}_{k,h}(s, \widetilde{\mathbf{a}}_{k,h}(s)) \leq \varepsilon_{k,h}. \tag{60}$$

Define $V_{k,h}(s)$ to be the clipped score achieved by this action, and use these values in the normalized transition regressions. Clipping is nondecreasing and one-Lipschitz, so it preserves the score-error bound. Membership in the candidate set preserves safety.

For $\chi_k = 0$, let $\mathbf{a}^\star = \Pi_h^\star(s)$ at a comparator-reachable state. Confidence and (60) give

$$\widetilde{Q}_{k,h}(s, \widetilde{\mathbf{a}}_{k,h}(s)) \geq r_h(s, \mathbf{a}^\star) + P_h V_{k,h+1}(s, \mathbf{a}^\star) - \varepsilon_{k,h}.$$

The sum $r_h + P_h V_{k,h+1}$ lies in $[0, H - h + 1]$, so clipping the left side preserves this lower bound. Subtracting it from the comparator Bellman recursion gives

$$V_h^{\Pi^\star}(s) - V_{k,h}(s) \leq \varepsilon_{k,h} + P_h(V_{h+1}^{\Pi^\star} - V_{k,h+1})(s, \Pi_h^\star(s)) \tag{61}$$

at comparator-reachable states. Backward induction bounds this difference by $\sum_{t=h}^{H} \varepsilon_{k,t}$. The optimism and Bellman-error argument for the actually executed protocol is unchanged. Its selected action defines its optimistic value at each state, and the induction does not require that action to be an exact maximizer. Hence the general regret bound gains at most

$$\sum_{k=1}^{K} \sum_{h=1}^{H} \varepsilon_{k,h}. \tag{62}$$

This accumulated planning error is at most $KH\varepsilon$ for a uniform error bound ε. The output must remain certified feasible because reward-regret control alone does not ensure safety.

C.6 Role of instantaneous constraints in the regret proof

Predictable targets and uniform queries. The historical transition targets in (10) use value functions fixed before their next-state observations. They therefore estimate the same parameter θ_h^P from predictable features, as in value-targeted regression [10]. Lemma C.1 and Cauchy–Schwarz imply, on one parameter-confidence event, that

$$\left|\widehat{P}_{k,h} V(s, \mathbf{a}) - P_h V(s, \mathbf{a})\right| \leq H \beta_{k,h}^P \|\psi_h^V(s, \mathbf{a})/H\|_{(\Lambda_{k,h}^P)^{-1}}. \tag{63}$$

This inequality holds simultaneously for all query features, including those formed from current data-dependent values. The factor H restores return units after normalization. In contrast, LSVI recomputes historical targets using the current value function and needs uniform concentration over a value-function class [40, full version]. Data-dependent safe sets also change that class, so a covering of action values alone is insufficient [6]. Our parameter argument applies to the specified historical-target estimator, not to arbitrary constrained LSVI.

Optimism and certification are distinct. Reward and transition bonuses compensate estimation errors. Constraint upper bounds restrict feasibility rather than penalize reward. On $\mathcal{E}$, the unclipped backup satisfies

$$0 \leq \widetilde{Q}_{k,h} - r_h - P_h V_{k,h+1} \leq 2b_{k,h}^r + 2b_{k,h}^P. \tag{64}$$

All terms use the same state–action arguments. Comparator optimism additionally requires its actions to be available throughout its reachable support. Otherwise, the analysis charges at most H for the episode through HB_K. Extra reward optimism cannot create missing safety observations, as Proposition C.8 demonstrates. The safe-subgraph bonuses in Shi et al. [5] rely on different support and geometric assumptions and do not automatically establish our coverage condition.

Counting and concentration. Lemma C.5 bounds executed-action widths using the features actually inserted into each design matrix. An unexecuted comparator feature cannot be charged to this count without additional information. Assumption 5.3 provides the observed covariance growth needed for that separate charge, while Proposition 5.6 counts actual observations of local table entries. Hoeffding's lemma bounds the normalized transition noise, self-normalized concentration controls adaptive regressions [36], and Azuma–Hoeffding controls initial-state and transition sampling errors. The trajectory martingale is defined on all histories, and its concentration event is then intersected with the parameter-confidence event. The analysis uses range-based bonuses. A Bernstein refinement would require variance estimation and its own error analysis [9], so no minimax-optimal horizon dependence is claimed.

D Proof of Theorem 5.5 (Necessity Lower Bounds)

These one-state, one-step constructions separate unstructured joint-action complexity, missing initial safety information, and statistical uncertainty near the constraint boundary.

Theorem D.1 (Unstructured joint-action lower bound)**.** *For $A \geq 2$, $K \geq A^N$, and any learning algorithm, there is a one-state, one-step cooperative game with N agents, A actions per agent, all joint actions safe, and expected regret at least $c\sqrt{A^N K}$, with the universal choice $c = (16\sqrt{2})^{-1}$.*

Proof of Theorem D.1. Use one global reward factor and let $M = A^N \geq 2$. Identify the joint actions with the arms of an M-armed stochastic bandit. In instance $j \in [M]$, arm j has Bernoulli mean $1/2 + \epsilon$ and the other arms have Bernoulli mean $1/2$, so $\epsilon > 0$ is the optimal arm's reward advantage. Let instance 0 have all means $1/2$, and write $N_j(K)$ for the number of plays of arm j. Let $\mathbb{P}_j$ be the law of the complete interaction on instance j, including the learner's randomization, and let $\mathbb{E}_j$ denote expectation under this law. Write KL for relative entropy and kl for its Bernoulli specialization. The chain rule and $\mathrm{kl}(1/2, 1/2 + \epsilon) \leq 4\epsilon^2$ for $0 < \epsilon \leq 1/4$ give

$$\mathrm{KL}(\mathbb{P}_0 \| \mathbb{P}_j) \leq 4\epsilon^2 \mathbb{E}_0 N_j(K).$$

Since $N_j(K) \in [0, K]$, Pinsker's inequality implies

$$\mathbb{E}_j N_j(K) \leq \mathbb{E}_0 N_j(K) + K\sqrt{2}\epsilon\sqrt{\mathbb{E}_0 N_j(K)}.$$

Summing over j and using $\sum_j N_j(K) = K$ yields

$$\frac{1}{M}\sum\nolimits_{j=1}^{M} \mathbb{E}_j \mathrm{Reg}(K) \geq \epsilon K \left(1 - \frac{1}{M} - \sqrt{2}\epsilon\sqrt{K/M}\right).$$

Choose $\epsilon = (4\sqrt{2})^{-1}\sqrt{M/K} \leq 1/4$. The bracket is at least $1/4$, so the average regret is at least $(16\sqrt{2})^{-1}\sqrt{MK}$. At least one instance has this regret. The safety constraints may be identically -1, so the lower bound applies even when safety is completely known. It concerns the unstructured reward class and does not establish the same dependence for a fixed-dimensional structured model. □

Theorem D.2 (Initial information is necessary for stringent safety)**.** *Without any initial safety certificate or safety information before the first action, some instance has first-action violation probability at least $1/2$. Thus a uniform $1 - \delta$ zero-violation guarantee is impossible for $\delta < 1/2$.*

Proof of Theorem D.2. Consider two one-state, one-step instances with two actions and identical information before the first action. In the first instance, only action one is safe, and in the second, only action two is safe. If the algorithm chooses action one with probability p, its violation probabilities in the two instances are $1 - p$ and p. Their maximum is at least $1/2$. Both instances admit a safe action. The impossibility is caused by the absence of information identifying which action is safe. □

Theorem D.3 (Safety-margin dependence of nontrivial certification)**.** *Let $Y_1, \ldots, Y_n$ be independent $\mathcal{N}(\theta, 1)$ observations, where $\theta \in \{-\gamma, +\gamma\}$ and $\gamma > 0$. A candidate action has linear constraint $g(a) = \theta$. If a rule declares it safe with probability at least $1 - \delta$ when $\theta = -\gamma$ and at most δ when $\theta = +\gamma$, where $0 < \delta \leq 1/4$, then*

$$n \geq \frac{\mathrm{kl}(1 - \delta, \delta)}{2\gamma^2} = \Omega\left(\frac{\log(1/\delta)}{\gamma^2}\right).$$

The observations may be given without executing the candidate.

Proof of Theorem D.3. Let $\mathbb{P}_-$ and $\mathbb{P}_+$ be the laws of the observations under $-\gamma$ and $+\gamma$. The Gaussian relative entropy is

$$\mathrm{KL}(\mathbb{P}_- \| \mathbb{P}_+) = 2n\gamma^2.$$

Let C be the event that the rule declares the candidate safe. Data processing gives

$$2n\gamma^2 \geq \mathrm{kl}(\mathbb{P}_-(C), \mathbb{P}_+(C)) \geq \mathrm{kl}(1-\delta, \delta).$$

The stipulated probability bounds imply the last step by monotonicity of binary relative entropy. Finally, $\mathrm{kl}(1-\delta,\delta) = (1-2\delta)\log((1-\delta)/\delta)$ has order $\log(1/\delta)$ when $0 < \delta \leq 1/4$.

This is a lower bound for simultaneous soundness and successful certification. Without the latter condition, always refusing to certify is a zero-sample rule. Because the construction already grants safe measurements, the bound also applies when the learner receives less information. □

E Planning and Distributed Coordination

E.1 Factorization of the complete optimistic score

The linear model and the planning factorization impose different requirements. Linear realizability controls statistical estimation but does not imply that an optimistic Bellman score has small action scopes. For example, the transition-value term $P_h V(s, \mathbf{a})$ can depend on all actions even when the immediate reward decomposes locally. The norm in a transition confidence bonus can introduce additional couplings. We therefore require a factorization of the complete, unclipped score,

$$\widetilde{Q}_{k,h}(s, \mathbf{a}) = \sum\nolimits_{f \in \mathcal{F}^Q_{k,h}} q_{k,h,f}(s_{J_f}, \mathbf{a}_{I_f}), \tag{65}$$

Here $\mathcal{F}^Q_{k,h}$ indexes the real-valued score factors $q_{k,h,f}$, whose action and state scopes are I_f and J_f. For any agent sets $I, J \subseteq [N]$, write

$$\mathcal{A}_I := \prod\nolimits_{i \in I} \mathcal{A}_i, \qquad \mathrm{proj}_J \, \mathcal{S} := \{s_J : s \in \mathcal{S}\}. \tag{66}$$

These are the local action alphabet and the possible local state inputs. An empty scope has the single empty tuple as its input. Thus factor f is evaluated on $\mathrm{proj}_{J_f} \mathcal{S} \times \mathcal{A}_{I_f}$. The local tables in this representation must be available to the planner. Their construction, including evaluation of transition-value features, is part of the computational cost. In particular, evaluating a feature such as $\psi^V_h(s, \mathbf{a}) = \sum_{s'} \phi^P_h(s' \mid s, \mathbf{a}) V(s')$ may require summation over the next-state space.

The score in (65) is the quantity optimized before value clipping. For a fixed state and step, the map $z \mapsto \max\{0, \min\{H - h + 1, z\}\}$ is nondecreasing. Consequently, a maximizer of $\widetilde{Q}_{k,h}$ is also a maximizer of the clipped score. The clipped function itself need not have an additive factorization.

Each immediate safety constraint defines a hard feasibility factor

$$c_{k,\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell}) = \begin{cases} 0, & z_\ell \in \mathcal{Z}^0_{\ell,h} \text{ or } U^g_{k,\ell,h}(z_\ell) \leq 0, \\ -\infty, & \text{otherwise.} \end{cases} \tag{67}$$

Here $z_\ell = (s_{I_\ell}, \mathbf{a}_{I_\ell})$, so initial and statistical certificates enter the same local feasibility factor. Its value $-\infty$ excludes an uncertified tuple from the maximization. Maximizing the sum of the score factors and these feasibility factors solves the optimization over $\widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s)$ whenever this set is nonempty. To evaluate the known fallback separately, compute $\widetilde{Q}_{k,h}(s, \mathbf{a}^0_h(s))$ and compare it with the certified optimum. This comparison implements optimization over $\widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s) \cup \{\mathbf{a}^0_h(s)\}$ without introducing a global disjunctive constraint into the factor graph.

State inputs and the physical communication graph. The action scopes I_f define the primal graph used for elimination. The state scopes J_f determine which observations are needed to instantiate a factor. A junction tree groups action variables into bags and specifies logical separator messages. The physical graph G_{comm} instead specifies one-round communication links between agents. Bags have computing hosts, and logical messages can be routed through agents outside their action scopes. These objects need not coincide. Hard safety factors use $J_f = I_f = I_\ell$, whereas score factors may use different state and action scopes.

For binary local states and a binary action a_1, consider the score

$$q(s, \mathbf{a}) = \mathbf{1}\{a_1 = s_1 \oplus \cdots \oplus s_N\} \tag{68}$$

This factor has action width zero and state scope $[N]$. Here $\oplus$ denotes addition modulo two, so its right-hand argument is the parity of the local state bits. It can arise as the first-step Bellman term of a known two-step model. The initial reward is zero, agent 1's next state equals the displayed indicator, and the other next local states are fixed. The terminal reward is agent 1's local state. All actions may be safe. A parity summary can avoid transmitting the raw global state to every agent, but it must convey information from all N inputs. Action width alone does not ensure local state access.

A direct encoding of one realized global state uses at most $\sum_i \lceil \log_2 |\mathcal{S}_i| \rceil$ bits before routing and headers. A tabular value function instead has $|\mathcal{S}|$ entries. An exponential state-space cardinality does not by itself imply exponential communication of one realized state.

E.2 An exact protocol for overlapping scopes

Write $\mathcal{N}_i^{\mathrm{comm}} := \{j : \{i, j\} \in E_{\mathrm{comm}}\}$ for agent i's physical neighbors. With integer round budget $0 \le R_{k,h} \le R_h$, a deterministic protocol at episode k and step h uses the message and action maps

$$M^t_{i\to j,k,h} = \mu^t_{i\to j,k,h}\left(s_i, \{M^u_{v\to i,k,h} : v \in \mathcal{N}_i^{\mathrm{comm}},\ u < t\}\right), \qquad \{i, j\} \in E_{\mathrm{comm}}, \tag{69}$$

$$a_i = \pi_{i,k,h}\left(s_i, \{M^u_{v\to i,k,h} : v \in \mathcal{N}_i^{\mathrm{comm}},\ u \le R_{k,h}\}\right). \tag{70}$$

Here $M^t_{i\to j,k,h}$ is the round-t message and $\mathbf{a} = (a_i)_{i\in[N]} = \Pi^k_h(s)$ is the final joint output. The maps and episode-specific model information are fixed using the training history before the episode. Their evaluation during execution uses only the local state and permitted messages. Thus $\Pi^k_h(s)$ is the maps' joint output, computed without a global-state policy-table lookup.

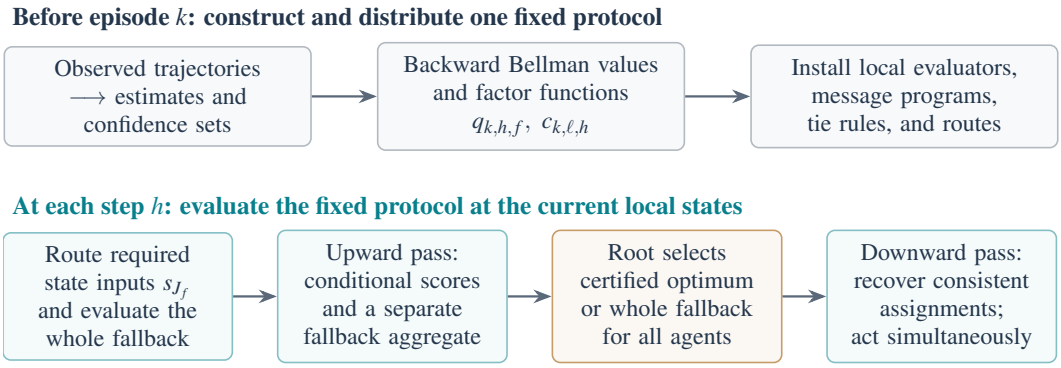


Figure 3: Execution of a fixed certified protocol. Pre-episode training fixes the model estimates, value functions, certificate maps, and factor programs. At the realized state, agents route local state and fallback inputs to their designated hosts. The upward pass aggregates conditional certified scores and the whole fallback's separate score. The root makes one common branch choice. The downward pass recovers consistent shared actions, which are delivered before simultaneous execution. Evaluating the current-state factors uses the fixed episode maps and does not refit the model.

Exact variable elimination solves the factorized optimization through conditional maximizations over action variables [37]. Exact implementability is a condition on one state-contingent protocol over the entire state space. Selecting a feasible action separately at each global state is insufficient if the resulting selections cannot be implemented from the agents' information. A sufficient implementation uses a junction tree of the full action-factor graph, together with routing over G_{comm} that permits each factor to be instantiated and every required message to be delivered before execution.

Let $\mathcal{T} = (\mathcal{B}, \mathcal{E}_{\mathcal{T}})$ be a rooted junction tree. Each bag $b \in \mathcal{B}$ contains a set B_b of action variables, every factor scope lies in at least one bag, and bags containing any given variable form a connected

subtree. Let b_0 be the root, $p(b)$ the parent of a nonroot bag, and $\mathrm{ch}(b) := \{c : p(c) = b\}$ its children. Assign each score or feasibility factor to exactly one bag containing its scope. At the current state, let $F_b(\mathbf{a}_{B_b})$ denote the sum of the factors assigned to bag b. For a nonroot bag b with parent $p(b)$, write $S_b = B_b \cap B_{p(b)}$ for its separator. Its upward message is the conditional subtree optimum,

$$m_{b\to p(b)}(\mathbf{a}_{S_b}) = \max_{\mathbf{a}_{B_b\setminus S_b}} \left\{ F_b(\mathbf{a}_{B_b}) + \sum\nolimits_{c\in\mathrm{ch}(b)} m_{c\to b}(\mathbf{a}_{S_c}) \right\}. \tag{71}$$

For each separator assignment, the bag stores a maximizing assignment of its remaining variables. Fix a total order on each finite action alphabet and use the increasing agent-index order within each bag to select the lexicographically first maximizing residual assignment. The root uses the same convention for its bag assignment. These choices and fallback priority on a score tie define the induced deterministic tie rule. Each upward message also carries the sum of the *score* factors evaluated at the whole fallback in its subtree, excluding hard feasibility factors from this scalar. The root obtains the certified optimum and the fallback score before selecting either branch, with fallback priority on a tie. An empty certified set has value $-\infty$ and therefore selects the fallback. The root sends one common branch flag down the tree. On the certified branch it also sends its separator assignments. Each child retrieves the maximizing assignment conditioned on the received separator and continues the pass. The running-intersection property ensures that each shared action variable has one value in every bag containing it. Output bags deliver each action and the common branch flag before the prescribed execution time.

Overlapping safety scopes thus share the same action variable throughout the optimization. They cannot select incompatible actions for a shared agent. The downward assignment pass is essential because independently selecting local maximizers can yield inconsistent actions under ties. A unilateral switch to a fallback component can violate a coupled constraint, even when both whole joint assignments are safe. The protocol assumes reliable, synchronous delivery and does not provide fault-tolerant commitment. Loopy max-sum alone does not guarantee exact optimization.

The order of elimination and message passing specifies computational dependencies. All agents share the team objective, and no follower best-response map is imposed. With exact optimization, the order can change runtime and the selected maximizer under ties, but not the optimal value or feasible set. This differs from a Stackelberg game, where the commitment order is part of the strategic model. Theorem 5.2 applies to every such exact, implementable planner. Corollary 5.4 additionally requires its coverage condition. Different tie choices can produce different observations and realized regret.

Example of shared-action consistency and branch selection. Consider binary actions with certified constraints $a_1 = a_2$ and $a_2 = a_3$, score factors $(1 - a_1)/3$ and $2a_3/3$, and fallback 000. Assign the first factor and constraint to bag $B_L = \{1, 2\}$ and the second pair to bag $B_R = \{2, 3\}$. Independent maximization selects $a_2 = 0$ in the left bag and $a_2 = 1$ in the right bag. Instead, root the junction tree at B_L. The right bag sends the score row $m_{R\to L}$ and its fallback score 0, while storing the conditional actions $a_3^\star$ locally as follows,

a_2	0	1
$m_{R\to L}(a_2)$	0	$2/3$
$a_3^\star(a_2)$	0	1

(72)

The root compares total scores $1/3$ for $a_1 = a_2 = 0$ and $2/3$ for $a_1 = a_2 = 1$, together with the fallback score $1/3$. It selects the certified branch and sends $a_2 = 1$ downward, recovering 111. If the physical graph is $1 - 2 - 3$ and the bags are hosted at agents 1 and 3, a logical tree message traverses two physical edges, and the chosen component must also reach agent 2. Both 000 and 111 are safe, whereas a componentwise mixture such as 101 is unsafe. Thus provisional local choices and individual fallback switches cannot replace the common branch and conditional assignment.

E.3 Cost of action selection and state-dependent planning

Let $\mathcal{F}$ contain all score and feasibility factors in the current planning problem. The primal action graph joins any two agents whose actions occur in the same factor. Let ω be the induced width of the selected elimination order, $A_{\max} = \max_i |\mathcal{A}_i|$, and $r_{\max} = \max_{f\in\mathcal{F}} |I_f|$. The induced width is the largest number of remaining neighbors of an eliminated variable after fill edges are added. Minimizing it over elimination orders gives treewidth. Local table construction takes at most $\sum_{f\in\mathcal{F}} A_{\max}^{|I_f|}$ factor

evaluations. After those tables are available, exact variable elimination requires

$$O\big(\mathrm{poly}(N, |\mathcal{F}|) A_{\max}^{\omega+1}\big) \tag{73}$$

arithmetic operations per state. This bound includes the scopes of transition values and confidence bonuses. The immediate safety graph alone is insufficient to determine ω.

A reduced junction tree with bags of size at most $\omega + 1$ has separator messages with at most $A_{\max}^{\omega}$ scalar entries. An upward pass and downward assignment pass use $O(|\mathcal{B}|)$ logical messages and $O(\mathrm{diam}(\mathcal{T}))$ synchronous junction-tree rounds. These counts assume that each logical tree edge can transmit its message in one round. Realizing them over G_{comm} adds routing latency and congestion. Gathering the state variables in J_f, constructing local tables, and agreeing on the fallback comparison also require permitted communication. These table bounds count scalar entries, rather than bits. Appendix I.2 gives bit-traffic and edge-capacity bounds for certified score quantization.

The computational benefit therefore concerns action optimization at a fixed state. Tabular backward planning evaluates $H|\mathcal{S}|$ state-step pairs, in addition to the cost of constructing transition-value features. Since $\mathcal{S}$ can contain the full product $\prod_i \mathcal{S}_i$, this state-space cost can itself be exponential in N. An implementation may instantiate the current state's action factors and solve its action optimization only when that state is visited. This avoids storing unused action tables, but does not remove the need to obtain the future value functions used in the Bellman score. The present analysis does not establish efficient planning over an arbitrarily large factored state space.

Small scopes alone do not ensure small induced width. For example, all-to-all pairwise factors have scopes of size two but induce a complete primal graph with $\omega = N - 1$. A factor containing all action variables also gives a complete primal graph. In these cases the generic elimination bound is exponential in N. Other graphs with many edges can have different widths, so the relevant parameter is the chosen induced width. Special algebraic structure may permit additional reductions, but is not implied by the scope-based representation used here.

E.4 Proof of Proposition E.1 (Planning Equivalence)

The pre-episode planner and the execution protocol operate on different inputs. Before episode k, the learner fixes its estimates, confidence sets, certificate maps, and the value functions $V_{k,h}$ through the backward recursion. It also fixes the complete score-factor maps, their hosting agents, routes, message schedule, and deterministic tie rules. At step h, the numeric tables $q_{k,h,f}(s^k_{J_f,h}, \cdot)$ and $c_{k,\ell,h}(s^k_{I_\ell,h}, \cdot)$ depend on the newly observed local states and are instantiated at their hosts. Upward maximization, fallback comparison, downward assignment, and output delivery are then computations performed during execution. Observations from this episode are stored for subsequent model updates and do not alter the maps used to make the current episode's decisions.

Define the immediate certified-set optimum and the whole-fallback score by

$$M^{\mathrm{imm}}_{k,h}(s) := \max_{\mathbf{a}\in\mathcal{A}} \left\{ \sum\nolimits_{f\in\mathcal{F}^Q_{k,h}} q_{k,h,f}(s_{J_f}, \mathbf{a}_{I_f}) + \sum\nolimits_{\ell=1}^{m} c_{k,\ell,h}(s_{I_\ell}, \mathbf{a}_{I_\ell}) \right\}, \tag{74}$$

$$M^0_{k,h}(s) := \widetilde{Q}_{k,h}(s, \mathbf{a}^0_h(s)). \tag{75}$$

The certified-set optimum is $-\infty$ when the immediate certified set is empty. Since the fallback is always a candidate and has finite score,

$$\begin{aligned} \max_{\mathbf{a}\in\widehat{\mathcal{A}}_{k,h}(s)} \widetilde{Q}_{k,h}(s,\mathbf{a}) &= \max\{M^{\mathrm{imm}}_{k,h}(s), M^0_{k,h}(s)\}, \\ V_{k,h}(s) &= \mathrm{clip}_{[0,H-h+1]}\big(\max\{M^{\mathrm{imm}}_{k,h}(s), M^0_{k,h}(s)\}\big). \end{aligned} \tag{76}$$

Thus the local constrained optimization, followed by a separate fallback comparison, implements the Bellman action set exactly. The known fallback is evaluated using only score factors and need not pass the statistical hard factors. If recursive continuation is imposed as in Appendix F, its test must also occur among the feasibility factors, with its full scope.

Proposition E.1 (Equivalence of precomputation and exact evaluation)**.** *Fix the training history before episode k. Suppose that the values, certificate maps, complete score factors, fallback, and coordination program of Proposition I.1 are fixed from that history at every step. Use the same induced deterministic tie rule in all evaluations. Then tabulating the program's output for every state*

and evaluating the program only at the realized states produce the same joint action and clipped score at every state. If the program satisfies the communication budget, its evaluation at realized states is a single admissible protocol and requires no centralized global-state table lookup. Exact caching of its factor tables also leaves its output unchanged. Caches are indexed by episode, step, factor, and the factor's local state input, or invalidated when the corresponding map changes.

Use the same rule for acquiring any auxiliary safety observations in both implementations. Consequently, coupling the initial states, transition randomness, and feedback in the two evaluation schedules gives identical trajectories and regression updates. The statistical guarantees are unchanged by this choice of evaluation schedule.

Proof of Proposition E.1. Conditional on the pre-episode history, every value function, factor map, route, and tie rule in the program is fixed. At any state s, a factor therefore has the same entries whether they are computed in advance, read from an exact cache, or evaluated when the state is encountered. The fallback protocol likewise induces the same component actions. Induction through the fixed upward message dependencies gives the same separator tables and fallback score at the root. The root comparison and the downward conditional assignments use the same tie rules, so they recover the same joint action and clipped maximum. The construction in Proposition I.1 expresses every message and output using permitted local information, without a centralized table lookup.

The equality holds for every state, including states not yet visited. With common initial-state and transition randomness, induction over environment steps gives the same states and actions. Coupling the feedback gives the same observations. Because the two schedules use the same fixed $V_{k,h+1}$, they also produce identical normalized transition features and responses in (10). Model updates at the end of the episode are therefore identical. Repeating this argument conditional on each subsequent training history proves the final claim. □

This equivalence requires exact evaluation of the fixed episode functions, including their successor-state dependencies. The feature $\psi_h^{V_{k,h+1}}(s, \mathbf{a})$ can require values at many next states. Memoization is valid only when all requested dependencies are evaluated and accessible. Arbitrary values at unvisited successors or model refitting during the episode would change the policy. Exact caching need not reduce the worst-case state-space computation. In particular, the finite-state implementation still computes $H|\mathcal{S}|$ value entries. An all-state joint-action table alone does not establish a decentralized lookup rule, because the agents may lack the information needed to identify its global-state row.

State scope, representation, and distribution cost. Action scopes I_f determine the induced width of elimination. State scopes J_f determine which current observations a factor host requires and need not equal I_f. Let $\mathcal{F}_{k,h}$ contain all score and hard feasibility factors at episode k and step h. If every factor is stored as a local table for all of its possible inputs, the number of entries is

$$\sum_{h=1}^{H} \sum_{f \in \mathcal{F}_{k,h}} |\operatorname{proj}_{J_f} \mathcal{S}| \prod_{i \in I_f} |\mathcal{A}_i|. \tag{77}$$

Hard safety factors have $J_f = I_f = I_\ell$, and fallback storage is counted separately. Formula (77) measures the tabular storage cost. A compact exact evaluator may use less storage, but its construction and internal evaluation cost must still be counted. Loading these maps or tables at their designated agents is pre-episode communication, excluded from the runtime round bound in Proposition I.1. Local action scopes do not by themselves bound storage or distribution costs. A transition-value factor can depend on the global state even with a single action input.

E.5 Sparse and dense forms of the motivating applications

The robotic and wireless examples illustrate the constraint structure. Their finite-state, finite-action versions fall within the formulation only when the stated feature and feedback assumptions also hold. At a fixed horizon, a common shift and positive scaling into $[0, 1]$ preserve policy ordering.

For collision avoidance, consider discretized robot states and controls and a constraint

$$g_{ij,h}(s_{\{i,j\}}, a_i, a_j) = d_{\min}^2 - \|p_{i,h}^+(s_i, a_i) - p_{j,h}^+(s_j, a_j)\|_2^2 \le 0. \tag{78}$$

Here $d_{\min} > 0$ is the required separation, and $p_{i,h}^+(s_i, a_i)$ is robot i's next-position vector associated with the local state and control. The constraint has pairwise action scope. It guarantees separation at

Phase	Computation and information	Cost accounting
Before episode k	Use $\mathcal{H}_{k-1}$ to fix estimates, certificates, and all Bellman values. Compile and distribute the factor evaluators, fallback maps, hosts, routes, schedule, and tie rules.	$H\|\mathcal{S}\|$ value entries in the tabular implementation. Factor construction, storage, and distribution are additional pre-episode costs.
Input acquisition at step h	Evaluate the fallback without acting. Route the current state components required by each factor and the fallback components needed for its score to the designated hosts.	At most $R_0 + r_{\mathrm{in}}$ rounds under the vector-message model. Payload and routing loads are separate resources.
Factor instantiation	Evaluate $q_{k,h,f}(s^k_{J_f,h}, \cdot)$ and the exact local feasibility indicators from the frozen episode maps.	At most $\sum_f A_{\max}^{\|I_f\|}$ factor evaluations. Internal evaluation cost is not included in this count.
Upward pass	Compute conditional maximum-score tables and stored maximizing assignments. Aggregate the whole fallback's score separately.	At most D_G rounds. Separator tables have at most $A^{\omega}_{\max}$ score entries.
Joint assignment and delivery	The root selects the certified candidate or the fallback. Propagate separator assignments or the common fallback flag, then deliver each selected component to its owner.	At most $D_G + r_{\mathrm{out}}$ rounds. All agents wait until the prescribed delivery phase has completed.
Environment action and feedback	Execute the committed components simultaneously and record rewards, constraint measurements, and the transition sample using the frozen $V_{k,h+1}$. Incorporate feedback into the next episode's estimates.	One environment step. The regret horizon counts these steps and excludes communication and computation time.

Table 4: Pre-episode planning and execution-time evaluation of one fixed certified protocol. Caching can replace exact table evaluations without changing the policy. It does not eliminate the need for future values or the communication needed to coordinate the current joint action.

the next sampled time only when the positions $p^+_{i,h}$ describe the actual next positions, or when a valid robust bound accounts for prediction error. Local collision constraints on a chain have bounded width if the optimistic value factors preserve that chain structure. Requiring pairwise collision checks for every robot pair instead gives a complete safety graph. A formation constraint involving all robots gives a global factor. These cases illustrate how coordination can remain expensive even though each physical action is local. A geometric-neighbor description alone does not guarantee bounded width.

For wireless resource allocation, let the finite action of transmitter i include a quantized power p_i. An interference constraint at receiver j with transmitter scope I_j may take the form

$$g_{j,h}(s_{I_j}, \mathbf{a}_{I_j}) = \sum\nolimits_{i \in I_j \setminus \{j\}} H_{ij,h} p_i - I_j^{\max} \le 0. \tag{79}$$

Here $H_{ij,h}$ is the interference-channel power gain from transmitter i to receiver j at step h, and $I_j^{\max}$ is its interference limit. Small interference neighborhoods can yield small factors, but a sparse implementation still requires bounded width of the complete planning graph. Network-wide interference or a shared total-power constraint can introduce a large factor. Some additive resource constraints admit specialized resource-indexed dynamic programs, so a large factor does not by itself establish intrinsic computational hardness. These examples specify constraint structures and their computational consequences. Controlled numerical evaluations appear in Appendix K.

F Backward Safety Certification and Robust Viability

The main analysis assumes a fallback that is safe at every modeled state and step. Hence, after any immediately safe action, at least one safe action is available at the next step. A transition-based viability test is unnecessary for this model. A different construction is required when the application specifies a terminal safe set, excludes states without safe actions, or restricts the fallback's domain.

For this extension, let $\mathcal{X}_{H+1} \subseteq \mathcal{S}$ be a given terminal safe set. An explicit immediate-certificate set is

$$\mathcal{A}^{\mathrm{cert}}_{k,h}(s) := \widehat{\mathcal{A}}^{\mathrm{imm}}_{k,h}(s), \tag{80}$$

using (14) with sound local initial certificates. Unlike the main candidate set, this set need not contain a fallback and can be empty. Known-safe pairs from a restricted fallback may be included in the initial certificates only where its immediate safety is known. The viability argument below also applies to any other immediately safe certificate set.

Define the admissible parameter intersection and its kernel confidence set by

$$\widehat{\Theta}^P_{k,h} := \left\{\theta \in \Theta^P_h : \|\theta - \widehat{\theta}^P_{k,h}\|_{\Lambda^P_{k,h}} \le \beta^P_{k,h}\right\},$$
$$\mathcal{C}^P_{k,h} := \begin{cases} \{P_\theta : \theta \in \widehat{\Theta}^P_{k,h}\}, & \widehat{\Theta}^P_{k,h} \neq \varnothing, \\ \{P_\theta : \theta \in \Theta^P_h\}, & \text{otherwise.} \end{cases} \tag{81}$$

Here P_θ is the transition kernel induced by parameter θ. The set Θ^P_h contains the valid-kernel parameters specified in Section 3.3. It is nonempty because it contains the true parameter. On the simultaneous confidence event, $\theta^P_h \in \widehat{\Theta}^P_{k,h}$ and $P_h \in \mathcal{C}^P_{k,h}$. The second branch makes the kernel set nonempty even when the intersection is empty off that event. This branch restores the prior kernel family without ensuring a safe recovery action or recursive feasibility. The certified state and action sets satisfy

$$\mathcal{X}_{k,H+1} = \mathcal{X}_{H+1}, \tag{82}$$
$$L^P_{k,h}(\mathcal{X}_{k,h+1} \mid s, \mathbf{a}) = \inf_{P \in \mathcal{C}^P_{k,h}} P(\mathcal{X}_{k,h+1} \mid s, \mathbf{a}), \tag{83}$$
$$\mathcal{A}^{\mathrm{rec}}_{k,h}(s) = \left\{\mathbf{a} \in \mathcal{A}^{\mathrm{cert}}_{k,h}(s) : L^P_{k,h}(\mathcal{X}_{k,h+1} \mid s, \mathbf{a}) = 1\right\}, \tag{84}$$
$$\mathcal{X}_{k,h} = \left\{s : \mathcal{A}^{\mathrm{rec}}_{k,h}(s) \neq \varnothing\right\}. \tag{85}$$

Here $L^P_{k,h}$ is the worst-case probability of reaching the next certified state set over the valid kernels in (81). The set $\mathcal{A}^{\mathrm{rec}}_{k,h}(s)$ collects immediately safe actions with probability-one certified continuation, and $\mathcal{X}_{k,h}$ contains states admitting such an action. The valid-kernel restriction ensures nonnegative, normalized transitions, so the infimum lies in $[0, 1]$.

The recursion removes immediately safe actions whose plausible successors lack a certified continuation. Inclusion of the true kernel transfers the robust transition condition to the actual dynamics.

Proposition F.1 (Robust viability)**.** *Suppose the true kernels belong to $\mathcal{C}^P_{k,h}$, every action in $\mathcal{A}^{\mathrm{cert}}_{k,h}(s)$ is immediately safe, and $\mathrm{supp}(\rho) \subseteq \mathcal{X}_{k,1}$. A protocol that selects an action in $\mathcal{A}^{\mathrm{rec}}_{k,h}(s_h)$ at every step satisfies all immediate constraints and remains in $\mathcal{X}_{k,h}$ at every step $h \in [H+1]$, almost surely. The assertion requires an implementable protocol. It does not establish a regret guarantee or provide an action when a recursive candidate set is empty.*

Proof of Proposition F.1. The initial state belongs to $\mathcal{X}_{k,1}$ almost surely. If $s_h \in \mathcal{X}_{k,h}$ and $\mathbf{a}_h \in \mathcal{A}^{\mathrm{rec}}_{k,h}(s_h)$, immediate safety follows from membership in $\mathcal{A}^{\mathrm{cert}}_{k,h}(s_h)$. Because the true kernel is one of the kernels in the confidence set,

$$P_h(\mathcal{X}_{k,h+1} \mid s_h, \mathbf{a}_h) \ge L^P_{k,h}(\mathcal{X}_{k,h+1} \mid s_h, \mathbf{a}_h) = 1.$$

Induction over h proves the claim. □

A fallback used in this extension must also satisfy the recursive continuation condition. Its immediate safety at a state does not imply that it avoids states without a safe next action. Feasibility from the initial distribution remains an additional assumption. In contrast, if $\mathcal{X}_{H+1} = \mathcal{S}$ and the globally safe fallback of the main model is included at every state, then $\mathcal{X}_{k,h} = \mathcal{S}$ for every h. This follows by backward induction because every valid kernel assigns probability one to $\mathcal{S}$. The continuation condition then imposes no additional restriction.

If the condition in (84) is relaxed to $L^P_{k,h}(\mathcal{X}_{k,h+1} \mid s, \mathbf{a}) \ge 1 - \eta$ for a specified $\eta \in [0, 1]$, each accepted action has true escape probability at most η while the certificates are valid. Suppose the initial-feasibility condition $\mathrm{supp}(\rho) \subseteq \mathcal{X}_{k,1}$ holds in every episode, and let δ_{conf} bound failure of the joint immediate-safety and transition-confidence event. A first-exit union bound over at most KH actions gives escape or confidence-failure probability at most $\delta_{\mathrm{conf}} + KH\eta$. This argument uses the conditional transition bound at each history with valid current certificates, without conditioning

future transitions on the entire confidence event. Even without a safe recovery action outside the recursively certified sets, zero violation therefore has probability at least $1 - \delta_{\mathrm{conf}} - KH\eta$. For a prescribed exit-error budget $\delta_{\mathrm{exit}} \in [0, 1]$, the explicit choice

$$\eta := \delta_{\mathrm{exit}}/(KH) \tag{86}$$

gives failure probability at most $\delta_{\mathrm{conf}} + \delta_{\mathrm{exit}}$. For example, the main confidence allocation permits $\delta_{\mathrm{conf}} = \delta/2$ and $\delta_{\mathrm{exit}} = \delta/2$. All feasibility and implementability conditions still apply. A positive tolerance cannot be treated as almost-sure invariance.

The strict threshold one requires certifying that the escape probability is exactly zero, which finite observations need not establish. For example, if the model family permits arbitrarily small positive probabilities of reaching an excluded state, observing only safe successors can leave such kernels in the confidence set after every finite sample sequence. The corresponding lower bound on safe-transition probability remains below one. A positive margin for the instantaneous constraint functions does not resolve this support-identification problem.

The immediate-constraint certification bound in the main theorem therefore does not imply a finite certification bound for this extension. Such a bound requires additional known transition-support information or a separate assumption ensuring that the comparator's continuation tests become certified. Any continuation test used in the action optimization must also be represented among its feasibility factors. Its scope contributes to the induced width and may be global even when all immediate safety constraints are local.

G Observation Models and Scope of the Guarantees

G.1 Aggregate reward feedback

Separate local reward regressions require the factor measurements in Assumption 3.1. If the learner instead observes only a scalar team reward Y^{agg} with conditional mean $r_h(s, \mathbf{a})$ and conditionally R_{agg}-sub-Gaussian noise, where R_{agg} is a given noise bound, it can use a joint regression. The stacked feature and parameter vectors in $\mathbb{R}^{D_r}$ are

$$\Phi_h^r(s, \mathbf{a}) = \big(\phi_{e,h}^r(s_e, \mathbf{a}_e)\big)_{e \in \mathcal{E}_r}, \qquad \Theta_h^r = \big(\theta_{e,h}^r\big)_{e \in \mathcal{E}_r}. \tag{87}$$

Stacking the $F \geq 1$ blocks gives $r_h = \langle \Phi_h^r, \Theta_h^r \rangle$, $\|\Phi_h^r\|_2 \leq \sqrt{F}$, and $\|\Theta_h^r\|_2 \leq \sqrt{F} B_r$. For each past episode $\tau < k$, the normalized feature and response are

$$x_{r,h}^{\mathrm{agg},\tau} = \Phi_h^r(s_h^\tau, \mathbf{a}_h^\tau)/\sqrt{F}, \qquad Y_{r,h}^{\mathrm{agg},\tau} = Y_h^{\mathrm{agg},\tau}/\sqrt{F}, \tag{88}$$

where $Y_h^{\mathrm{agg},\tau}$ is the observed, unnormalized team-reward measurement. The feature has norm at most one, the conditional mean response is $\langle x_{r,h}^{\mathrm{agg},\tau}, \Theta_h^r \rangle$, and the normalized noise scale is $R_{\mathrm{agg}}/\sqrt{F}$. Applying (31) gives the joint design, estimate, and confidence radius

$$\begin{aligned} \Lambda_{k,h}^{\mathrm{agg}} &= I_{D_r} + \sum\nolimits_{\tau<k} x_{r,h}^{\mathrm{agg},\tau} (x_{r,h}^{\mathrm{agg},\tau})^\top, \\ \widehat{\Theta}_{k,h}^{r,\mathrm{agg}} &= (\Lambda_{k,h}^{\mathrm{agg}})^{-1} \sum\nolimits_{\tau<k} x_{r,h}^{\mathrm{agg},\tau} Y_{r,h}^{\mathrm{agg},\tau}, \\ \beta_{k,h}^{\mathrm{agg}} &= \beta(\Lambda_{k,h}^{\mathrm{agg}}; \sqrt{F} B_r, R_{\mathrm{agg}}/\sqrt{F}). \end{aligned} \tag{89}$$

Use the same failure allocation α as in (32). Replacing F reward regressions by one makes this allocation conservative. The unnormalized reward estimate and prediction bonus are

$$\begin{aligned} \widehat{r}_{k,h}^{\mathrm{agg}}(s, \mathbf{a}) &= \langle \Phi_h^r(s, \mathbf{a}), \widehat{\Theta}_{k,h}^{r,\mathrm{agg}} \rangle, \\ b_{k,h}^{\mathrm{agg}}(s, \mathbf{a}) &= \sqrt{F}\, \beta_{k,h}^{\mathrm{agg}} \|\Phi_h^r(s, \mathbf{a})/\sqrt{F}\|_{(\Lambda_{k,h}^{\mathrm{agg}})^{-1}}. \end{aligned} \tag{90}$$

Replacing the reward estimate and bonus throughout the algorithm yields the same safety proof and regret decomposition. Write $\beta^{\mathrm{agg}} = \max_{k \in [K], h \in [H]} \beta_{k,h}^{\mathrm{agg}}$ for the largest joint-regression radius. Its reward-uncertainty contribution to regret is at most

$$2\sqrt{F}\, \beta^{\mathrm{agg}} \sqrt{2HD_rT \log(1 + K/D_r)}. \tag{91}$$

This follows by applying Lemma C.5's determinant argument to the normalized joint features at each step. The other terms are unchanged. The joint confidence norm generally couples all reward scopes, so aggregate feedback can require larger planning factors than factorwise feedback.

G.2 Scope of the guarantees

The main model assumes a known safe fallback at every modeled state and step. This condition is stronger than knowing a policy that is safe only along its trajectories from the initial distribution and excludes irreversible dead ends. The optional recursion in Appendix F addresses viability on a smaller domain, but requires feasible initial states and transition-support certificates. An immediate safety margin alone does not imply the main finite certification bound for that extension.

The sublinear-regret corollary additionally assumes that safely obtained data reduce uncertainty in relevant comparator feature directions. The existence of a baseline and strict safety margins on initially uncertified comparator tuples does not imply this coverage. Initial local certificates can cover boundary tuples, but their soundness must be established independently of noisy point estimates. Auxiliary safety observations, when available, must enter the constraint designs and sample counts. Their acquisition cost is absent from the environment regret. The result therefore does not establish sample-efficient safe exploration under every trajectory-only feedback model.

Exact decentralized execution requires sufficient reliable pre-action communication to reproduce the certified joint assignment. Proposition I.2 accounts for finite score-message precision and gives a sufficient schedule under edge capacities, conditional on correctly computed feasibility indicators. It does not provide a finite-precision implementation of arbitrary-real regression and safety tests. Packet losses, asynchronous execution, adversarial participants, and the effect of communication time on reward are outside the model. Independent local fallback switches after a communication failure can violate coupled constraints. Proposition H.1 shows that even treewidth one does not ensure implementability under a short round budget.

Computational scalability additionally depends on state-input acquisition, trajectory collection for centralized training, and distribution of the learned factor programs. The state space, program representation, factor evaluations, and routing loads can be large despite bounded action width. These costs are separate from the per-state action-optimization bound, which does not establish scalable planning for arbitrary factored state spaces.

The reward, constraint, and transition models must be correctly specified. Safety is asserted for the deterministic functions $g_{\ell,h}$, whose noisy measurements are used for estimation. The result does not constrain every realization of an unrelated random measurement or disturbance. Extending the analysis to misspecified or nonlinear models, partial observations that do not support the required coordination, and competitive objectives remains open. Appendix K evaluates safe learning on fixed-width chains and exact planning under sparse and dense coupling. These synthetic results do not establish performance in physical systems or under model misspecification.

Broader impacts. Learning coordinated actions subject to instantaneous constraints could help reduce unsafe behavior in robotic teams and limit interference in wireless systems. These potential benefits depend on the validity of the modeled constraints, the availability of the assumed safe fallback and feedback, and the communication required to execute a certified joint action. Model misspecification or implementation failures can invalidate the guarantee and create unwarranted confidence in an unsafe controller. Moreover, maximizing a common team reward subject to safety constraints does not by itself ensure fair resource allocation. Applications require validation of the constraint model and execution mechanism, together with any additional fairness or privacy requirements relevant to the deployment.

H Proofs of Proposition H.1 and Corollary H.2

A centralized MDP representation preserves the joint dynamics and team reward. It does not, by itself, enforce the information available to each executing agent. The following example separates the optimum over communication-compatible safe protocols from the optimum of a full-state controller, even when statistical estimation and large factor scopes are absent. The argument uses causal locality of synchronous communication [41].

Proposition H.1 (A sharp communication requirement for hard-safe coordination)**.** *For every $N \geq 2$, there is a one-step cooperative game with two global states, binary actions, a unary reward factor, pairwise safety factors, and a known safe fallback at every state, with the following properties. All model functions can be revealed to the agents, every feasible action has safety margin one, and the*

complete reward–constraint action graph has treewidth one. Let $V_R^\star$ be the optimal value among safe protocols using at most R synchronous communication rounds on a path, and let $V_{\mathrm{full}}^\star$ be the optimal value of a full-state safe controller. Then

$$V_{\mathrm{full}}^\star = 1, \qquad V_R^\star = \begin{cases} \frac{1}{2}, & 0 \le R < N-1, \\ 1, & R \ge N-1. \end{cases} \tag{92}$$

The lower bound permits arbitrary message sizes and shared or private randomness independent of the current state. More generally, every $R < N-1$ protocol, whether safe or not, satisfies

$$\mathbb{E}[r(s,\mathbf{a})] \le \tfrac{1}{2} + \mathbb{P}(\exists \ell : g_\ell(s_{I_\ell}, \mathbf{a}_{I_\ell}) > 0)\,. \tag{93}$$

Equality constraints force the remote endpoint to share the informed agent's action. Until a message can traverse the path, that endpoint's decision cannot depend on the reward-relevant bit. The resulting loss of safe value persists even when the model is completely known.

Proof of Proposition H.1. Let G_{comm} be the path $1 - 2 - \cdots - N$. The only random local state is $s_1 = X$, where $X \sim \mathrm{Bernoulli}(1/2)$, and s_i is fixed for every $i > 1$. The bit is drawn after the episode's shared training information has been fixed. Each action belongs to $\{0,1\}$, and

$$r(s,\mathbf{a}) = \mathbf{1}\{a_1 = X\}, \qquad g_i(a_i, a_{i+1}) = 2\mathbf{1}\{a_i \ne a_{i+1}\} - 1, \quad i \in [N-1]. \tag{94}$$

The constraints require the common action $a_1 = \cdots = a_N$. Their graph is a path, and every safe joint action has $g_i = -1$ on every edge. The all-zero protocol is a known fallback safe in both states. The reward is a single local factor at agent 1. Both the reward and each constraint have a one-dimensional linear representation with feature equal to the corresponding displayed function and coefficient one. All feature magnitudes are at most one. The transition after the single decision can be any fixed known kernel on the two states and does not affect the value.

Condition on the shared training information and all random seeds. After zero communication rounds, an agent's information depends only on its own local state. Inductively, after t rounds, its information can depend only on local states within graph distance t, because a round-t message is a function of the sender's local state and its earlier transcript, and traverses one edge. Thus, if $R < N-1$, agent N cannot distinguish the two values of X. Its action a_N is independent of X, also after averaging over the state-independent seeds. Consequently, $\mathbb{P}(a_N = X) = 1/2$.

If $a_1 \ne a_N$, at least one adjacent pair has unequal actions, so

$$\mathbf{1}\{a_1 = X\} \le \mathbf{1}\{a_N = X\} + \mathbf{1}\{\exists i \in [N-1] : a_i \ne a_{i+1}\}. \tag{95}$$

Taking expectations proves (93). An almost-surely safe protocol therefore has value at most $1/2$, and the all-zero fallback attains this value.

A full-state controller chooses $a_i = X$ for every agent, obtaining reward one and satisfying every constraint. This choice is also implementable in $N-1$ rounds. Agent 1 sends X to agent 2, and each subsequent round forwards the bit one edge farther along the path. Afterwards all agents choose the bit they received. This proves (92). Finally, without the coupling constraints, agent 1 could obtain reward one with no communication by setting $a_1 = X$. The round requirement in the example arises from coordinating the safe joint action. □

Corollary H.2 (Centralized regret under a communication limit)**.** *Consider K independent episodes of* (94)*, with a fresh fair bit in each episode and at most $R < N-1$ rounds per decision. The protocols may depend on the entire preceding training history. If the probability of any constraint violation during the K episodes is at most δ, then*

$$K V_{\mathrm{full}}^\star - \mathbb{E}\left[\sum\nolimits_{k=1}^{K} r(s^k, \mathbf{a}^k)\right] \ge K\big(\tfrac{1}{2} - \delta\big). \tag{96}$$

Proof of Corollary H.2. Let E_k be the event of an edge-constraint violation in episode k. Conditional on the previous training history, the new bit is independent and fair, so the proof of (93) applies unchanged. Taking expectations and summing gives $\mathbb{E}\sum_k r(s^k,\mathbf{a}^k) \le K/2 + \sum_k \mathbb{P}(E_k)$. Since each $E_k \subseteq \bigcup_{j=1}^K E_j$, the last sum is at most $K\delta$. Use $V_{\mathrm{full}}^\star = 1$. □

Equation (96) compares against the unrestricted full-state controller. For $R < N - 1$, the admissible comparator in (4) has value $1/2$, which the fallback already attains. The full-state maximizing rule is then unimplementable, so Assumption 3.3 fails. This example isolates the communication requirement in that assumption. Bounded action-graph treewidth alone does not bound the rounds needed to attain the full-state safe optimum.

I Proofs of Distributed Planning and Communication Guarantees

I.1 Proof of Proposition I.1 (Distributed Implementation)

The following construction gives verifiable sufficient conditions for Assumption 3.3. Exact elimination and distributed action coordination in factored cooperative models are established techniques [7, 8, 37]. Here the construction applies them to the complete certified planning problem and accounts separately for state acquisition, the safe fallback, and delivery of the selected actions.

Fix an episode and a step, and suppress their indices. Suppose the complete unclipped score in (16) has known factors

$$\widetilde{Q}(s, \mathbf{a}) = \sum\nolimits_{f \in \mathcal{F}^Q} q_f(s_{J_f}, \mathbf{a}_{I_f}), \tag{97}$$

with finite values. Here $\mathcal{F}^Q$ indexes score factors, and I_f, J_f specify their action and state inputs. Allow sound initial local certificates $\mathcal{Z}^0_{\ell,h}$, meaning that $g_{\ell,h}(z) \le 0$ for every $z \in \mathcal{Z}^0_{\ell,h}$. For the fixed episode and step, define the local feasibility indicator

$$c_\ell(z) = \begin{cases} 0, & z \in \mathcal{Z}^0_{\ell,h} \text{ or } U^g_{k,\ell,h}(z) \le 0, \\ -\infty, & \text{otherwise,} \end{cases} \qquad z = (s_{I_\ell}, \mathbf{a}_{I_\ell}). \tag{98}$$

The immediate certified set is the intersection of these local tests. Each disjunction is evaluated within its original constraint scope and adds no new action dependency. Empty initial certificate sets recover the statistical tests alone. Let $\mathcal{F}$ contain all score factors and the hard factors (98). For a hard factor associated with constraint ℓ, set $J_f = I_f = I_\ell$. All factor maps and their scopes are fixed by the pre-episode training information. The relevant maps, including the membership rule for each $\mathcal{Z}^0_{\ell,h}$, are available at their designated computing agents. Their state arguments are acquired during execution. Write $G = G_{\mathrm{comm}}$ for the undirected physical graph. Each directed edge can carry an arbitrary finite vector per synchronous round. Local computations and routing instructions are deterministic. Communication is measured in rounds here. Bit traffic and packet sizes are accounted for separately below. The class $\mathfrak{P}_R(G)$ contains the deterministic protocols in (2) using this message model and at most R rounds on G.

Let $\mathcal{T}$ be a fixed rooted, reduced junction tree, meaning that no bag is contained in an adjacent bag, with bags $\{B_b : b \in \mathcal{B}\}$ of size at most $\omega + 1$. The bags cover all action variables, contain every factor's action scope, and satisfy the running-intersection property. Each bag b is hosted by an agent u_b. Assign each factor f to one bag $b(f)$ containing I_f, and each action variable i to one output bag $b(i)$ containing i. Thus $u_b \in [N]$ is a computing agent, $b(f)$ is the factor's bag, and $b(i)$ is the bag responsible for delivering agent i's selected action. Let b_0 be the root, $p(b)$ the parent of nonroot bag b, and $\mathrm{ch}(b) := \{c : p(c) = b\}$ its children. Use fixed shortest physical paths for this round construction. Ties between shortest paths are resolved before observing the current state. The bag assignments, hosts, action-alphabet orders, and paths are all part of the installed program. The later bit-traffic accounting also allows other specified simple routes and uses their actual lengths.

Write d_G for graph distance. Define the maximum input-routing distance

$$\begin{aligned} r_{\mathrm{in}} = \max\big(\{0\} \cup \{d_G(j, u_{b(f)}) : f \in \mathcal{F},\ j \in J_f\} \\ \cup \{d_G(i, u_{b(f)}) : f \in \mathcal{F}^Q,\ i \in I_f\}\big), \end{aligned} \tag{99}$$

The second line accounts for the fallback actions needed to evaluate its score. For root b_0, define the routed tree height and output distance by

$$\begin{aligned} D_G &= \max_{b \in \mathcal{B}} \sum_{(c, p(c)) \in \mathrm{path}_{\mathcal{T}}(b, b_0)} d_G(u_c, u_{p(c)}), \\ r_{\mathrm{out}} &= \max_{i \in [N]} d_G(u_{b(i)}, i). \end{aligned} \tag{100}$$

All distances used here must be finite. The quantity D_G is the largest sum of routed edge lengths along a logical root-to-bag path, while r_{out} is the maximum distance from an output host to its action owner. In particular, D_G need not equal the physical diameter. A known fallback protocol computes its component actions within a given, state-uniform integer budget $R_0 \geq 0$. Its maps can be evaluated without acting, with completion padded to R_0 rounds.

Proposition I.1 (Compilation of a certified planner into a protocol)**.** *Under the preceding conditions, if the permitted round budget satisfies*

$$R \;\geq\; R_0 + r_{\mathrm{in}} + 2D_G + r_{\mathrm{out}}, \tag{101}$$

there is a single deterministic state-contingent protocol in $\mathfrak{P}_R(G)$ that returns, for every state s,

$$\mathbf{a}(s) \in \arg \max_{\mathbf{a} \in \widehat{\mathcal{A}}^{\mathrm{imm}}(s) \cup \{\mathbf{a}^0(s)\}} \widetilde{Q}(s, \mathbf{a}). \tag{102}$$

The protocol has a fixed tie rule and also computes the exact clipped maximum in (17)*. It needs at most $\sum_{f \in \mathcal{F}} A_{\max}^{|I_f|}$ factor-table evaluations, followed by $O((|\mathcal{F}| + |\mathcal{B}|)A_{\max}^{\omega+1} + N)$ arithmetic operations for optimization at a fixed state. Each upward junction-tree message contains at most $A_{\max}^{\omega}$ score entries and one additional scalar for the fallback score. Each direction uses $|\mathcal{B}| - 1$ logical messages. Input acquisition, output delivery, and physical routing are counted separately.*

If the conditions hold at every step of the backward recursion and the permitted protocol class allows all local message maps with the stated graph, alphabet, and step-specific round budgets, this construction verifies Assumption 3.3 without a runtime global-state policy-table lookup.

Proof of Proposition I.1. First evaluate the fallback's message and action maps, without acting in the environment, in R_0 rounds. Agent i now knows $a_i^0(s)$. Send each state component required by a factor to that factor's host, and send each fallback component required by a score factor to its host. Use fixed shortest paths with tagged messages. Because an edge may transmit an arbitrary finite vector, packets sharing an edge can be concatenated, and all these transmissions finish in at most r_{in} rounds. Each bag host then evaluates its score and feasibility tables, together with the score of the whole fallback for its assigned factors.

At bag b, let $F_b(\mathbf{a}_{B_b})$ be the sum of all its assigned score and hard feasibility factors. The bag's fallback score is

$$Q_b^0(s) = \sum_{f \in \mathcal{F}^Q:\, b(f)=b} q_f(s_{J_f}, \mathbf{a}_{I_f}^0(s)). \tag{103}$$

Only score factors occur in (103), because the known fallback need not pass the statistical upper-confidence tests. For a nonroot bag with separator $S_b = B_b \cap B_{p(b)}$, the upward message is

$$m_{b \to p(b)}(\mathbf{a}_{S_b}) = \max_{\mathbf{a}_{B_b \setminus S_b}} \left[F_b(\mathbf{a}_{B_b}) + \sum\nolimits_{c \in \mathrm{ch}(b)} m_{c \to b}(\mathbf{a}_{S_c}) \right]. \tag{104}$$

Store a maximizing assignment for each separator value, using a fixed ordering of the bag variables and their actions to resolve ties as in Appendix E. Each upward message also carries the subtree fallback score

$$Q_{b\uparrow}^0(s) := Q_b^0(s) + \sum\nolimits_{c \in \mathrm{ch}(b)} Q_{c\uparrow}^0(s). \tag{105}$$

At a leaf the sum is zero. The root also evaluates the two branch scores

$$\begin{aligned} M_{\mathrm{cert}}(s) &:= \max_{\mathbf{a}_{B_{b_0}} \in \mathcal{A}_{B_{b_0}}} \left[F_{b_0}(\mathbf{a}_{B_{b_0}}) + \sum\nolimits_{c \in \mathrm{ch}(b_0)} m_{c \to b_0}(\mathbf{a}_{S_c}) \right], \\ M_0(s) &:= Q_{b_0\uparrow}^0(s) = \widetilde{Q}(s, \mathbf{a}^0(s)). \end{aligned} \tag{106}$$

The scores M_{cert} and M_0 compare the certified branch with the whole fallback. The root's product action alphabet $\mathcal{A}_{B_{b_0}}$ is defined in (66).

The running-intersection property implies that a variable occurring in the subtree of b and outside that subtree must occur in S_b. Therefore, conditional on $\mathbf{a}_{S_b}$, all other variables in that subtree can be maximized internally. Induction from the leaves proves that (104) is exactly the maximum sum of the subtree's factors subject to the separator assignment. The root's maximum $M_{\mathrm{cert}}(s)$ is consequently the exact optimum over $\widehat{\mathcal{A}}^{\mathrm{imm}}(s)$, taking $-\infty$ when empty. The fallback aggregate is $M_0(s)$.

The root selects the certified optimum when $M_{\mathrm{cert}}(s) > M_0(s)$ and otherwise selects the fallback. This fixes fallback priority on a tie. If the certified optimum is selected, the root chooses its maximizing bag assignment, then sends the corresponding separator assignments down the tree. Each child retrieves the stored maximizing assignment conditioned on the received separator and continues the pass. Since the root's selected value is finite, every visited conditional optimum is feasible. The running-intersection property makes every shared action variable consistent throughout the tree, and the recovered joint action satisfies every hard factor. On the confidence event, such an action satisfies every true constraint, since each accepted local tuple is initially certified or has a nonpositive valid upper bound. If the fallback is selected, the downward pass sends this common selection instead. Each output bag then sends the appropriate selection flag and, when needed, the selected action component to its owning agent. On the fallback branch, agent i uses its already computed $a_i^0(s)$, so its output bag need not acquire that component to deliver the selection flag. All agents execute after final delivery. Both branches satisfy (102), and monotone clipping gives the Bellman value in (17).

Each logical tree edge is routed along its fixed shortest physical path. An upward message is sent after all of its child messages have arrived. These dependency times can be fixed in advance from the routing lengths. The longest root-to-leaf sum of those lengths is D_G, so the upward pass finishes in at most D_G rounds. The downward pass obeys the reverse dependency order and takes at most another D_G rounds. Concurrent routed messages do not incur a capacity constraint under the stated vector-message model. Final delivery takes at most r_{out} rounds, proving (101).

Every message is a deterministic function of the hosting agent's local state, fixed pre-episode information, and previously received messages. All table evaluations, routes, schedules, and tie rules are specified independently of the realized state. Their composition is therefore one protocol over the entire state space, rather than an unrelated collection of statewise assignments. The computed optimum can also be evaluated for every state during finite-state backward planning. This establishes the full value and protocol requirement in Assumption 3.3 when the conditions hold at each step.

There are at most $A_{\max}^{|I_f|}$ entries to evaluate in factor f. Once these tables are available, summing factors in the bags and forming all messages takes $O((|\mathcal{F}| + |\mathcal{B}|)A_{\max}^{\omega+1} + N)$ arithmetic operations, including recovery of the N output components. A reduced junction tree with maximal bag size $\omega + 1$ has separators of size at most ω. Thus the upward table and message counts have the asserted sizes. Downward messages contain a selection flag and one separator assignment. □

The sufficient round budget can be reduced when additional structure permits a shorter protocol, as in the direct bit broadcast of Proposition H.1. Its applicability still requires factorization of the complete score and access to each factor's state arguments. Local safety scopes and linear realizability alone provide neither condition. The factor-table evaluation count excludes each factor's internal evaluation cost, such as summing transition features over next states. Finite-state backward planning also retains its $H|\mathcal{S}|$ state–step cost. Accounting for finite score messages additionally requires the state-input demands, payloads, and physical routes specified below.

I.2 Proof of Proposition I.2 (Communication Precision)

The round bound in Proposition I.1 assumes unbounded vector payloads. The following accounting separates local state acquisition from separator messages and shows the effect of communicating scores at a prescribed precision. All state and action alphabets are finite. Let b_j^s and b_i^a be the fixed bit lengths for encoding agent j's state and agent i's action,

$$b_j^s = \lceil \log_2 |\mathcal{S}_j| \rceil, \qquad b_i^a = \lceil \log_2 |\mathcal{A}_i| \rceil. \tag{107}$$

For a hard factor associated with constraint ℓ, set $J_f = I_f = I_\ell$. The state-input demands $\mathcal{D}_s$ and fallback-action demands $\mathcal{D}_0$ are the source–host pairs

$$\begin{aligned} \mathcal{D}_s &= \{(j, u_{b(f)}) : f \in \mathcal{F},\ j \in J_f\}, \\ \mathcal{D}_0 &= \{(i, u_{b(f)}) : f \in \mathcal{F}^Q,\ i \in I_f\}. \end{aligned} \tag{108}$$

Repeated requests for the same component at the same host are merged. Send each requested component from its source to its host along a fixed simple path. Write $\mathcal{R}_{j\to u}$ for a state-input route. The state-input traffic, summed over traversed directed edges, is

$$W_s = \sum\nolimits_{(j,u)\in\mathcal{D}_s} |\mathcal{R}_{j\to u}| b_j^s. \tag{109}$$

This quantity depends on the state scopes and routing, not only on action treewidth. A factor with a small action scope can still require many state components. Conversely, local state scopes and local hosts can avoid broadcasting the entire state to every agent. The message protocol imposes no state-privacy guarantee, since a factor table or separator message can reveal information about states outside the recipient's local observation.

Proposition I.2 (Score precision and certified feasibility)**.** *Fix an episode and step satisfying the structural and access conditions of Proposition I.1. Suppose the hard-factor indicators are evaluated correctly. A state-uniform score bound is*

$$L_f := \max_{\substack{u \in \mathrm{proj}_{J_f} \mathcal{S} \\ v \in \mathcal{A}_{I_f}}} |q_f(u, v)|, \qquad f \in \mathcal{F}^Q. \tag{110}$$

The local alphabets are finite and the score maps have finite values, so $0 \le L_f < \infty$. This quantity depends only on the frozen episode maps, not on the realized state. An explicitly established larger bound may be used instead. Computing the displayed maximum is part of factor preprocessing. For a precision $\zeta > 0$ fixed before the current state is observed, replace each score factor in action selection by its quantized integer score

$$z_f(s_{J_f}, \mathbf{a}_{I_f}) = \left\lfloor q_f(s_{J_f}, \mathbf{a}_{I_f})/\zeta \right\rfloor, \qquad m_Q = |\mathcal{F}^Q|. \tag{111}$$

Run exact integer max-sum on these factors, retain the same exact hard factors, and compare the certified optimum with the fallback under the same integer objective. The returned action $\widehat{\mathbf{a}}(s)$ belongs to $\widehat{\mathcal{A}}(s)$ for every state and satisfies

$$0 \le \max_{\mathbf{a} \in \widehat{\mathcal{A}}(s)} \widetilde{Q}(s, \mathbf{a}) - \widetilde{Q}(s, \widehat{\mathbf{a}}(s)) \le m_Q \zeta. \tag{112}$$

To bound score encodings and table sizes, define

$$Z_\zeta = \sum\nolimits_{f \in \mathcal{F}^Q} \lceil L_f/\zeta \rceil, \qquad b_\zeta = \lceil \log_2(2Z_\zeta + 2) \rceil, \qquad T_b = \prod\nolimits_{i \in S_b} |\mathcal{A}_i|. \tag{113}$$

Here Z_ζ bounds the magnitude of any integer partial score sum, b_ζ is a uniform codeword length, and T_b is the number of separator assignments for nonroot bag b. The factor count m_Q controls the total rounding loss. Every upward table entry and fallback-score aggregate can be encoded in b_ζ bits, including an infeasibility symbol. The execution messages other than those internal to the fallback protocol admit the following fixed payload bounds shown below.

Message	*Source and destination*	*Payload in bits*
State input	$(j, u) \in \mathcal{D}_s$	b_j^s
Fallback action	$(i, u) \in \mathcal{D}_0$	b_i^a
Upward table and fallback sum	$u_b \to u_{p(b)}$	$(T_b + 1)b_\zeta$
Downward commitment	$u_{p(b)} \to u_b$	$1 + \sum_{i \in S_b} b_i^a$
Final action delivery	$u_{b(i)} \to i$	$1 + b_i^a$

Let $\mathcal{M}$ be this fixed list of messages, with payload length b_p and simple directed physical route $\mathcal{R}_p$ for message $p \in \mathcal{M}$. For a specified finite-bit fallback implementation, let $\mathrm{bits}_e^0(s)$ count the bits that it sends on directed physical edge e at state s. If these counts are finite, take $W_e^0 := \max_{s \in \mathcal{S}} \mathrm{bits}_e^0(s)$, or any explicitly established larger bound. Then the complete execution has per-edge load and total traffic bounded by

$$W_e = W_e^0 + \sum\nolimits_{p \in \mathcal{M}:\ e \in \mathcal{R}_p} b_p, \qquad W = \sum\nolimits_e W_e. \tag{114}$$

Fixed schedules and encodings identify packets implicitly. The bounds exclude transport headers.

Quantization is applied once to the original score factors. Intermediate messages are exact integers, so rounding loss accumulates across factors independently of the number of routing hops. The hard tests and the separate fallback branch are preserved, leaving the candidate set unchanged.

Proof of Proposition I.2. For every factor and input, $0 \le q_f - \zeta z_f < \zeta$. Summing the m_Q factors gives

$$0 \le \widetilde{Q}(s, \mathbf{a}) - \zeta \sum\nolimits_f z_f(s_{J_f}, \mathbf{a}_{I_f}) < m_Q \zeta \tag{115}$$

when $m_Q > 0$, with zero quantization error when $m_Q = 0$. Let $\mathbf{a}^\star$ maximize the unrounded score over the candidate set. The integer optimizer, including its fallback comparison, satisfies $\sum_f z_f(s, \mathbf{a}^\star) \le \sum_f z_f(s, \widehat{\mathbf{a}})$. Apply (115) to both actions to obtain (112). The feasible set did not change, so feasibility and the confidence-event safety guarantee are preserved.

Every integer local score has magnitude at most $\lceil L_f/\zeta \rceil$. Because each factor is assigned to exactly one bag, a feasible message entry is a maximum of sums of distinct factors in one subtree. Its magnitude is at most Z_ζ. An infeasible entry uses a separate symbol. The alphabet $\{-Z_\zeta, \dots, Z_\zeta\} \cup \{\perp\}$ therefore needs b_ζ bits. Fallback aggregates obey the same bound. There are T_b separator assignments, plus one fallback scalar. Downward messages carry the common branch flag and one separator assignment, while output messages carry that flag and one action component. Padding the fallback branch to the same lengths gives fixed payloads in both cases. Recipients know their fallback components and ignore padded action bits. Summing routed payloads gives (114). □

The precision ζ is distinct from the continuation tolerance η in Appendix F. Correct evaluation of a hard test such as $U^g \le 0$ may require exact or certified local arithmetic. An uncertain feasibility indicator must not be rounded into acceptance. The proposition therefore bounds finite-bit *communication* conditional on correct indicators. It does not establish a finite-precision implementation of their computation. The episode's model maps, certificate descriptions, routing schedules, and factor programs must already be installed at their hosts. Their storage and distribution cost is separate and need not have a finite fixed-precision representation under the main statistical model.

Regret consequence. For this approximate planner, compute each backward value from the original unrounded score of the selected action,

$$V_{k,h}(s) = \mathrm{clip}_{[0,H-h+1]}\big(\widetilde{Q}_{k,h}(s, \widehat{\mathbf{a}}_{k,h}(s))\big).$$

At every backward step, the factorization, factor bounds, routing conditions, and exact feasibility tests must hold for the score built from this planner's own achieved $V_{k,h+1}$. They are not inherited from a separate exact-maximization recursion. Freeze the resulting values before the episode and use them in the historical transition samples (10). Define $m_{Q,k,h} := |\mathcal{F}^Q_{k,h}|$, the number of complete score factors at that episode and step. Choose $\zeta_{k,h} > 0$ before observing the current state, uniformly over the state space. Appendix C.5 then adds at most

$$\sum_{k=1}^{K} \sum_{h=1}^{H} m_{Q,k,h} \zeta_{k,h} \tag{116}$$

to the existing regret bound, while preserving zero violation on the confidence event. For a known $T = KH$, choosing $\zeta_{k,h} = 1/[T \max\{1, m_{Q,k,h}\}]$ makes this additional term at most one. For this choice, $Z_\zeta \le m_Q + T \max\{1, m_Q\} \sum_f L_f$, giving an explicit logarithmic score-word bound through (113). Certification coverage must hold for the modified algorithm's actual feedback. Deterministic rounding adds no failure event, but can change the trajectories and numerical confidence widths. The backward values remain the clipped original scores achieved by the selected actions.

A sufficient schedule with edge capacities. If directed physical edge e can transmit a positive integer c_e bits per round, one valid, conservative implementation sends the listed messages one at a time, hop by hop, in the input, upward, downward, and output dependency order. If the fallback finishes in R_0 rounds under those same capacities, the whole protocol finishes within

$$R_0 + \sum_{p \in \mathcal{M}} \sum_{e \in \mathcal{R}_p} \lceil b_p / c_e \rceil \tag{117}$$

rounds. Serial transmission removes congestion between messages, while parallel use of different edges can yield shorter schedules. The permitted round budget must accommodate the schedule actually used. Both the original vector-message bound and this capacitated bound assume that the environment does not advance until joint commitment completes.

J Proofs of Local Certification and Feasible-Deviation Results

J.1 Proof of Lemma J.1 (Local Certificate Closure)

For constraint ℓ, write

$$\mathcal{Z}_\ell = \{(s_{I_\ell}, \mathbf{a}_{I_\ell}) : (s, \mathbf{a}) \in \mathcal{S} \times \mathcal{A}\}, \qquad z_\ell(s, \mathbf{a}) = (s_{I_\ell}, \mathbf{a}_{I_\ell}). \tag{118}$$

Thus $\mathcal{Z}_\ell$ is the finite configuration set for constraint ℓ, and $z_\ell(s, \mathbf{a})$ extracts its local state–action tuple. The initial local certificate set $\mathcal{Z}^0_{\ell,h}$ contains only tuples whose constraint values are known to be nonpositive. Given a collection $\mathcal{B}_h \subseteq \mathcal{S} \times \mathcal{A}$ of known-safe joint state-action pairs, one valid choice is

$$\mathcal{Z}^0_{\ell,h} = \{z_\ell(s, \mathbf{a}) : (s, \mathbf{a}) \in \mathcal{B}_h\}. \tag{119}$$

The global fallback provides the collection $\mathcal{B}_h = \{(s, \mathbf{a}^0_h(s)) : s \in \mathcal{S}\}$. Additional known-safe pairs may also be included. Because $g_{\ell,h}$ depends only on its local tuple, each projected certificate remains valid under every global completion of that tuple. Combining these projections can therefore certify additional joint actions whenever their shared variables are consistent. Constructing the local certificate sets from a global seed description incurs a separate preprocessing cost.

Lemma J.1 (Maximal safety information from seed projections)**.** *Fix the scopes and safe seeds $\mathcal{B}_h$, and use the projections in* (119)*. Define their consistent join*

$$\mathcal{J}_h(s) := \{\mathbf{a} : z_\ell(s, \mathbf{a}) \in \mathcal{Z}^0_{\ell,h} \text{ for every } \ell\}. \tag{120}$$

Every action in $\mathcal{J}_h(s)$ is safe under every collection of local constraint tables consistent with the seeds. If the tables are otherwise unrestricted, every action outside $\mathcal{J}_h(s)$ is unsafe for some collection of local tables consistent with the same seeds. Thus $\mathcal{J}_h(s)$ is exactly the set whose safety follows from the seed membership statements and the known scopes alone.

Proof of Lemma J.1. Every seed satisfies every constraint. Whenever a tuple occurs in a seed's projection for constraint ℓ, the value of that tuple under this constraint is nonpositive. Membership in $\mathcal{J}_h(s)$ provides such a certificate for every constraint on the same joint action, proving safety.

Conversely, let $\mathbf{a} \notin \mathcal{J}_h(s)$. There is a constraint ℓ for which $z = z_\ell(s, \mathbf{a}) \notin \mathcal{Z}^0_{\ell,h}$. Choose its table to equal $+1$ at z and -1 at every other configuration, and choose all other constraint tables to be constantly -1. The distinguished tuple occurs in no seed for this constraint. Every seed is therefore safe, but $(s, \mathbf{a})$ violates constraint ℓ. The values ± 1 can be replaced by any allowed common positive magnitude if bounded tables are required. This proves maximality in the unrestricted local table class. Additional known linear relations can imply further safe configurations. □

For example, at one state with scopes $\{1, 2\}$ and $\{2, 3\}$, the safe seeds 000 and 101 certify the local relations $\{00, 10\}$ and $\{00, 01\}$. Their consistent join contains $000, 001, 100, 101$. The recombined actions 001 and 100 require no additional constraint observations to establish their safety. An initial local certificate can likewise be combined with a statistical certificate from another scope. In each case, shared action variables must have the same values in every constraint containing them.

J.2 Example of certification at the safety boundary

Local certificates can admit an optimal action on one constraint's boundary while observations certify its remaining constraints with positive margins. To isolate this effect, compare local certification with treating the fallback as one indivisible candidate. Consider one state, one step, and three binary-action agents, with

$$\begin{aligned} r(\mathbf{a}) &= a_3, \qquad g_1(a_1, a_2) = |a_1 - a_2|, \\ g_2(a_2, a_3) &= 2a_2a_3 - 1, \qquad \mathbf{a}^0 = 000. \end{aligned} \tag{121}$$

The two constraint scopes are $I_1 = \{1, 2\}$ and $I_2 = \{2, 3\}$. We omit the single fixed state from feature and tuple arguments. Use the one-hot feature map for g_1, with entries ordered as $00, 01, 10, 11$ and true parameter $\theta^g_{1,1} = (0, 1, 1, 0)^\top$. For g_2, use $\phi^g_{2,1}(a_2, a_3) = (2a_2a_3 - 1, 0, 0, 0)^\top$ and $\theta^g_{2,1} = (1, 0, 0, 0)^\top$. Both models thus have dimension $d_g = 4$. Set $B_g = \sqrt{2}$. Measurements have zero noise, so $R_g = 0$ and every constraint radius in (32) is $\sqrt{2}$. The reward feature is a_3, with true parameter one, $B_r = 1$, and $R_r = 0$. All feature norms and parameter norms meet the model assumptions. The transition after the single decision has no effect on value. Regression data sets are initially empty, and the regularization is I.

The safe team optimum 001 has reward one, whereas the fallback has reward zero. Consider first the candidate-set rule

$$\{\mathbf{a} : U^g_{k,\ell,1}(z_\ell(s, \mathbf{a})) \le 0 \text{ for every } \ell\} \cup \{000\},$$

which uses the seed only as a complete joint action. This rule executes 000 forever. Indeed, after n such observations, the 00 estimate in the first constraint table is zero, and its upper confidence bound is $\sqrt{2}/\sqrt{1+n} > 0$. Each other entry remains unobserved, with upper bound $\sqrt{2} > 0$. Thus every joint action fails the first statistical test, leaving only the fallback. Regret is K.

Projecting the seed instead gives $\mathcal{Z}^0_{1,1} = \mathcal{Z}^0_{2,1} = \{00\}$. Its first local certificate applies to both 000 and 001, even though the first constraint holds with equality there. The fallback and the candidate have the same second-constraint feature, namely $(-1, 0, 0, 0)^\top$. After n fallback observations, the candidate's second upper confidence bound is

$$U^g_{k,2,1}(01) = -\frac{n}{1+n} + \frac{\sqrt{2}}{\sqrt{1+n}}.$$

This expression is positive for $n = 0, 1, 2$ and negative for every $n \geq 3$, since $n^2 > 2(1+n)$ holds exactly for integers $n \geq 3$. The local composition rule therefore selects 000 for the first three episodes and admits 001 at episode four. Before its first execution, the candidate's reward score is one, while the fallback's score is zero. After n_r observations of the candidate, its optimistic reward score is

$$\frac{n_r}{1+n_r} + \frac{1}{\sqrt{1+n_r}} \geq 1.$$

Every execution of 001 adds the same second-constraint feature, so its safety certificate remains valid. The planner consequently continues to select 001, with no tie between it and the fallback. Regret is exactly $\min\{K, 3\}$, and safety holds throughout.

The initially uncertified comparator tuple in the second constraint has margin one, and each fallback observation adds exactly its rank-one design increment. The margin and coverage conditions restricted to initially uncertified tuples therefore hold without auxiliary observations. The optimal tuple in the first constraint has zero margin but is initially certified. The restricted margin condition therefore covers this instance, whereas a positive-margin requirement on every constraint of every nonfallback comparator action would exclude it. A centralized method with the same local certificates can exploit this information. The two certification rules differ in how they represent and reuse the safe seed.

J.3 Proof of Proposition 5.6 (Local Certification)

Definitions and probability allocation. For each local constraint table, let $M_\ell = |\mathcal{Z}_\ell|$ and let $n_{k,\ell,h}(z)$ be the number of observations of entry z available before episode k. The observations have conditional mean $g_{\ell,h}(z)$ and conditionally R_g-sub-Gaussian noise. Their selection is predictable, and the known bound n_g limits the total number of observations for each pair (ℓ, h) through episode K. For positive count $n = n_{k,\ell,h}(z)$, let

$$\overline{g}_{k,\ell,h}(z) := n^{-1} \sum\nolimits_{j=1}^{n} Y^{(j)}_{\ell,h,z}, \tag{122}$$

where $Y^{(j)}_{\ell,h,z}$ is the jth observed value of entry z. Auxiliary observations contribute to this count only when their conditional mean is that entry's value. An arbitrary linear functional is not an entry observation. The scalar certificate uses

$$\alpha = \frac{\delta}{2H(F+m+1)}, \qquad L_\ell = \log\left(\frac{2M_\ell n_g}{\alpha}\right). \tag{123}$$

Here α allocates failure probability per process, and L_ℓ covers all entries and positive counts. For a positive count, define the tabular safety upper bound

$$U^{g,\mathrm{tab}}_{k,\ell,h}(z) = \overline{g}_{k,\ell,h}(z) + R_g\sqrt{\frac{2L_\ell}{n_{k,\ell,h}(z)}}.$$

Set $U^{g,\mathrm{tab}}_{k,\ell,h}(z) = +\infty$ at count zero. Initially certified tuples in $\mathcal{Z}^0_{\ell,h}$ require no observations.

Fix (ℓ, h, z). For each $n \in \{1, \ldots, n_g\}$, the conditional sub-Gaussian exponential bound gives

$$\mathbb{P}\left(\left|n^{-1}\sum\nolimits_{j=1}^{n} Y^{(j)}_{\ell,h,z} - g_{\ell,h}(z)\right| > R_g\sqrt{\frac{2L_\ell}{n}},\ \text{the entry is observed at least } n \text{ times}\right) \leq 2e^{-L_\ell}.$$

The sum uses the first n observations of this entry, even when observations of other entries occur between them. To justify the bound under adaptive sampling, let z_t be the entry selected at observation index t for the fixed pair (ℓ, h), and let $\widetilde{Y}_t$ be its response. Define its cumulative selected count and centered response sum by

$$N_t(z) := \sum_{v=1}^{t} \mathbf{1}\{z_v = z\}, \qquad S_t(z) := \sum_{v=1}^{t} \mathbf{1}\{z_v = z\}[\widetilde{Y}_v - g_{\ell,h}(z)]. \tag{124}$$

If fewer than n_g observations are acquired, extend the count and sum processes to n_g positions with zero increments. For each real tilting parameter u, conditional sub-Gaussianity and predictable selection make $\exp\{uS_t(z) - u^2 R_g^2 N_t(z)/2\}$ a nonnegative supermartingale with initial value one. Stop it when $N_t(z)$ first reaches n, or at the final observation bound n_g if this has not occurred. For $R_g > 0$, choose $u = \sqrt{2L_\ell/n}/R_g$ for the upper tail and its negative for the lower tail. On each corresponding deviation event at count n, the stopped exponential is at least e^{L_ℓ}. The supermartingale expectation bound therefore gives e^{-L_ℓ} for each tail and $2e^{-L_\ell}$ in total. If $R_g = 0$, observed responses equal their conditional means almost surely, so the claim is immediate.

A union bound over the M_ℓ entries and their at most n_g positive counts has failure probability at most $2M_\ell n_g e^{-L_\ell} = \alpha$. Thus, simultaneously for every episode-start data set,

$$|\overline{g}_{k,\ell,h}(z) - g_{\ell,h}(z)| \le R_g \sqrt{\frac{2L_\ell}{n_{k,\ell,h}(z)}} \quad \text{if } n_{k,\ell,h}(z) \ge 1. \tag{125}$$

There are Hm such table events. Retain the reward and transition parameter-confidence events with failure probability α for each of their $HF + H$ processes. The total confidence failure probability is at most $H(F + m + 1)\alpha = \delta/2$. Denote the combined event by $\mathcal{E}_{\mathrm{tab}}$.

On this event every statistical local certificate is sound. Unobserved entries require an initial certificate or the whole fallback. Every candidate satisfies the true constraints, so the modified sets retain the zero-violation guarantee.

Counting unavailable comparator episodes. Let

$$q_\ell = \max\left\{1, \left\lceil \frac{8R_g^2 L_\ell}{\Delta_{\mathrm{safe}}^2} \right\rceil \right\}.$$

The integer q_ℓ is a sufficient sample count used only in the analysis. Evaluating the statistical certificate requires neither this threshold nor the unknown margin Δ_{safe}. An action outside the candidate set is not the fallback and fails at least one local test at a tuple outside $\mathcal{Z}_{\ell,h}^0$. For a comparator action, this tuple has $g_{\ell,h}(z) \le -\Delta_{\mathrm{safe}}$ by the restricted margin condition. If $n_{k,\ell,h}(z) \ge q_\ell$, its count is positive, and (125) implies

$$U_{k,\ell,h}^{g,\mathrm{tab}}(z) \le g_{\ell,h}(z) + 2R_g \sqrt{\frac{2L_\ell}{n_{k,\ell,h}(z)}} \le -\Delta_{\mathrm{safe}} + \Delta_{\mathrm{safe}} = 0.$$

For $R_g = 0$, one observation suffices and the confidence term is zero. Every failed test at an initially uncertified comparator tuple therefore has $n_{k,\ell,h}(z) < q_\ell$.

Assign each episode with $\chi_k = 1$ to one witness (ℓ, h, z) whose count increases by at least one during that episode, as required by the proposition's coverage condition. The count never decreases. A fixed triple can be assigned at most q_ℓ episodes, because its pre-episode counts at those assignments are distinct integers in $\{0, \ldots, q_\ell - 1\}$. Only initially uncertified entries can be assigned. Hence

$$B_K \le \sum_{h=1}^{H} \sum_{\ell=1}^{m} |\mathcal{Z}_\ell \setminus \mathcal{Z}_{\ell,h}^0|\, q_\ell \le H \sum_{\ell=1}^{m} M_\ell q_\ell. \tag{126}$$

For one-hot table observations, let e_z be the coordinate unit vector of $\mathbb{R}^{M_\ell}$ indexed by z, and write $n_{k,\ell,h} = (n_{k,\ell,h}(z))_{z \in \mathcal{Z}_\ell}$ for the count vector. The design is $I + \mathrm{diag}(n_{k,\ell,h})$. Its increment dominates $e_z e_z^\top$ precisely when the count of entry z increases by at least one. Thus the count condition is the one-hot specialization of the covariance-growth condition in (25). The counting argument therefore requires actual observations of each charged entry.

Regret. The proof of Lemma C.3 requires sound candidate sets and availability of comparator actions on their reachable support. It does not require ellipsoidal safety bounds. The regret decomposition and the reward and transition potential bounds are also unchanged. Intersect $\mathcal{E}_{\mathrm{tab}}$ with the trajectory concentration event of Lemma C.4, whose failure probability is $\delta/2$. The intersection has probability at least $1-\delta$. On that event the zero-violation and regret statements hold jointly, and substituting (126) gives

$$\begin{aligned}\mathrm{Reg}(K) \le 2\beta^r\sqrt{2FHD_rT\log(1+K)} + 2H\beta^P\sqrt{2Hd_PT\log(1+K/d_P)} \\ + H\sum_{h=1}^{H}\sum_{\ell=1}^{m}|\mathcal{Z}_\ell \setminus \mathcal{Z}^0_{\ell,h}|\,q_\ell + H\sqrt{2T\log(2/\delta)}.\end{aligned}$$

Replacing the middle sum by $H\sum_\ell M_\ell q_\ell$ yields the stated bound. Every auxiliary safety observation contributes to these counts, while regret measures only the loss in environment returns.

Local representation size. The product form of the joint action space gives

$$M_\ell = |\operatorname{proj}_{I_\ell}\mathcal{S}|\prod_{i\in I_\ell}|\mathcal{A}_i|.$$

Define the maximum local state-space size $S_{\max} := \max_i|\mathcal{S}_i|$ and $r_g := \max(\{0\}\cup\{|I_\ell| : \ell \in [m]\})$, the largest constraint-scope size. Since $|\mathcal{A}_i| \le A_{\max}$, $\sum_\ell M_\ell \le m(S_{\max}A_{\max})^{r_g}$. The certification count is therefore governed by distinct local configurations. For a one-state binary-action chain with pairwise constraints, there are $4H(N-1)$ local entries across all steps, compared with $H(N-1)2^N$ entries in an unstructured table of each constraint over full joint actions. These are representation and certification counts. They do not bound state-space planning, transition representation, or induced action-graph width.

J.4 Proof of Corollary 5.7 (Feasible Deviations)

For a safe protocol Π, the feasible unilateral-deviation class $\mathfrak{D}_i(\Pi)$ contains the identity replacement and consists only of protocols in $\mathfrak{P}_{\mathrm{safe}}$. In particular, feasibility applies to the whole trajectory induced by the deviated message and action maps. The quantity $\mathrm{Gap}_i(\Pi)$ is the supremum of the common-return improvement over this class. The identity replacement gives $\mathrm{Gap}_i(\Pi)\ge 0$. Every $\widetilde{\Pi}\in\mathfrak{D}_i(\Pi)$ satisfies

$$V_1^{\widetilde{\Pi}}(\rho) - V_1^{\Pi}(\rho) \le V_1^{\Pi^\star}(\rho) - V_1^{\Pi}(\rho),$$

because $\Pi^\star$ maximizes the same common payoff over $\mathfrak{P}_{\mathrm{safe}}$. Taking the supremum over deviations, then the maximum over agents, proves the pointwise gap inequality.

On the simultaneous confidence and regret event, soundness of every candidate set ensures that each learned protocol Π^k belongs to $\mathfrak{P}_{\mathrm{safe}}$. Applying the pointwise inequality to each such protocol and summing gives

$$\sum_{k=1}^{K}\max_i \mathrm{Gap}_i(\Pi^k) \le \sum_{k=1}^{K}\left[V_1^{\Pi^\star}(\rho) - V_1^{\Pi^k}(\rho)\right] = \mathrm{Reg}(K).$$

The inequality holds on the same joint event. Consequently, the average maximum feasible-deviation gap is at most $\mathrm{Reg}(K)/K$. For any $\epsilon>0$, at most $\mathrm{Reg}(K)/\epsilon$ episodes have maximum gap greater than ϵ. These gaps concern safe admissible protocol deviations and values from ρ. Equilibrium conditions at off-path states are not implied.

Feasible equilibria can have strictly suboptimal team value. Consider one state, one step, two binary-action agents, and no pre-action messages. Let the common reward and constraint be

$$r(a_1,a_2) = \frac{a_1+a_2}{2}, \qquad g(a_1,a_2) = |a_1-a_2| - \frac{1}{2}.$$

The safe set is exactly $\{(0,0),(1,1)\}$, and both actions have safety margin $1/2$. The reward is the sum of two local factors, and the constraint has a one-hot linear realization. All model values can be known in advance, and $(0,0)$ is a globally safe fallback.

At $(0,0)$, changing either agent's action alone produces a mismatch and violates the constraint. Each feasible unilateral-deviation class therefore contains only the identity action, and both unilateral gaps are zero. The admissible team optimum $(1,1)$ has value one, while $(0,0)$ has value zero. Repeating $(0,0)$ gives zero cumulative feasible-deviation gap but team regret K. Thus small feasible unilateral gaps do not imply team optimality. Team optimality implies unilateral stability for the shared reward.

K Experiments on Safe Learning and Structured Planning

K.1 Safe learning at fixed action width

We isolate local safety learning on binary-action chains with $N \in \{4, 8, 16, 32, 64\}$, $H = 1$, and $K = 5000$ episodes. Additional numerical results and extended discussions are provided in the extended version [42]. This one-step specialization tests certificate reuse and agent-count scaling, rather than transition estimation or multistep planning. The global state space is $\mathcal{S} = \{(0, \dots, 0), (1, \dots, 1)\}$. Every agent observes the common mode $s \in \{0, 1\}$, drawn independently and uniformly in each episode. Action a_i selects power $p_s(a_i)$, with local reward and adjacent-pair constraint

$$\begin{aligned} p_s(a_i) &= p_s^0 + (1 - p_s^0)a_i, \\ r_i(s, a_i) &= w_{i,s} p_s(a_i)/N, \\ g_\ell(s, a_\ell, a_{\ell+1}) &= c_\ell \big[p_s(a_\ell) + p_s(a_{\ell+1})\big] - b_\ell, \qquad \ell \in [N-1]. \end{aligned} \tag{127}$$

Here $p_0^0 = 0.1$ and $p_1^0 = 0.5$ are low-power levels. The unknown gains c_ℓ, threshold ratios b_ℓ/c_ℓ, and reward weights $w_{i,s}$ are independent uniform draws from $[0.95, 1.05]$, $[1.72, 1.78]$, and $[0.4, 1]$, respectively. The team reward is in $[0, 1]$. Adjacent active actions violate the constraint, whereas the other three local action pairs are safe in both modes.

Reward features are $\phi_i^r = p_s(a_i)e_s$, where e_0, e_1 are the coordinate vectors of $\mathbb{R}^2$. Constraint features are $\phi_\ell^g = (p_s(a_\ell) + p_s(a_{\ell+1}), -1)^\top/\sqrt{5}$. Their parameters are $\theta_i^r = (w_{i,0}, w_{i,1})^\top/N$ and $\theta_\ell^g = \sqrt{5}(c_\ell, b_\ell)^\top$. We use (32) with $B_r = \sqrt{2}/N$, $B_g = 5$, $\lambda = 1$, and $\delta = 0.05$, without a tuned confidence multiplier. Each execution provides all factor rewards and constraints, with independent Gaussian noise of standard deviations $R_r = 0.05/N$ and $R_g = 0.02$. Initially, only all-low local tuples and the whole all-zero fallback are certified. The terminal value is zero, so transition-value terms vanish. Reward estimates and bonuses are unary, and pairwise safety constraints give complete action width one.

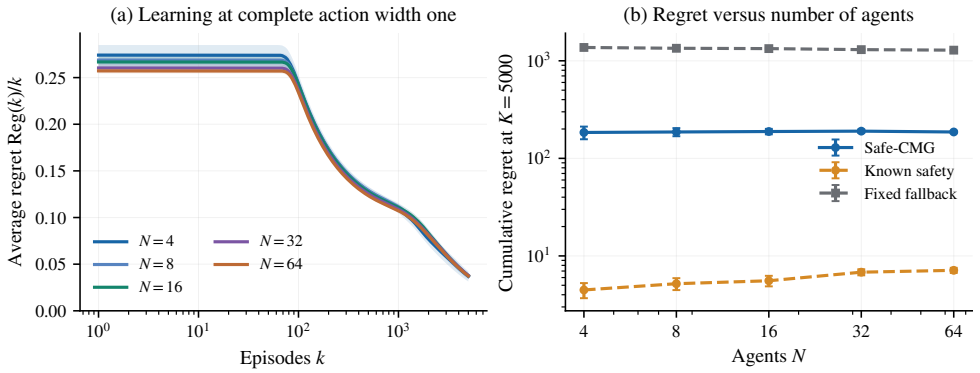


Figure 4: Safe learning at $H = 1$ and complete action width one. Left, average regret $\mathrm{Reg}(k)/k$. Right, cumulative regret at $K = 5000$ for learned safety, known safety, and the fixed fallback. Bands and bars are approximate pointwise 95% intervals over 30 seeds. Rewards are normalized by N, and each execution provides all local factor observations.

Exact suffix maximization and conditional assignment recovery enforce consistency across overlapping scopes, with lexicographic ties. The *known-safety* comparison uses the same reward estimator and planner with the true feasible sets. The all-zero fallback is a fixed safe reference. At every episode, the true comparator and frozen learned policy are evaluated exactly in both modes to compute (4). All 30 seeds (24001–24030) are included. Each seed uses a prefix of one 256-agent parameter draw across N, and methods use matched mode and noise streams. Distinct seeds are independent. Reported intervals are $\bar{y} \pm t_{29,0.975}\widehat{\sigma}/\sqrt{30}$, where $\bar{y}$ and $\widehat{\sigma}$ are the run mean and standard deviation and $t_{29,0.975}$ is the Student-t quantile.

Figure 4 shows cumulative regret of 184.3–190.1, compared with 1286.0–1370.4 for the fallback. All 17,850,000 executed scope–step tests in the learned-safety sweep satisfy the true constraints. This

illustrates safe improvement on the tested family, not an agent-count-independent worst-case rate. Reward normalization and increasing scalar feedback with N affect the comparison. Finite-window curves and zero observed violations do not establish asymptotic rates or high-probability guarantees.

K.2 Exact planning under sparse and dense coupling

We compare exact variable elimination and joint-action enumeration on identical score and feasibility tables. For binary actions, $\widetilde{Q}(\mathbf{a}) = \sum_i w_i a_i$, with independent integer weights uniform on $\{1, \ldots, 1000\}$. Each graph edge imposes $a_i a_j = 0$. Both planners use lexicographic ties and compare the whole all-zero fallback separately, preferring it on a score tie. Elimination uses descending agent order and conditional backtracking.

Chains use $N \in \{4, 8, 12, 16, 20, 32, 64, 128, 256\}$ and cliques use even N from 4 to 18, with 25 independent instances per size and master seed 20261105. Enumeration is run only for $N \leq 20$. Timings include validation, optimization, assignment recovery, and fallback comparison, but exclude factor construction. One independent warmup precedes each configuration, and method order alternates across instances. Figure 5 reports medians and interquartile ranges.

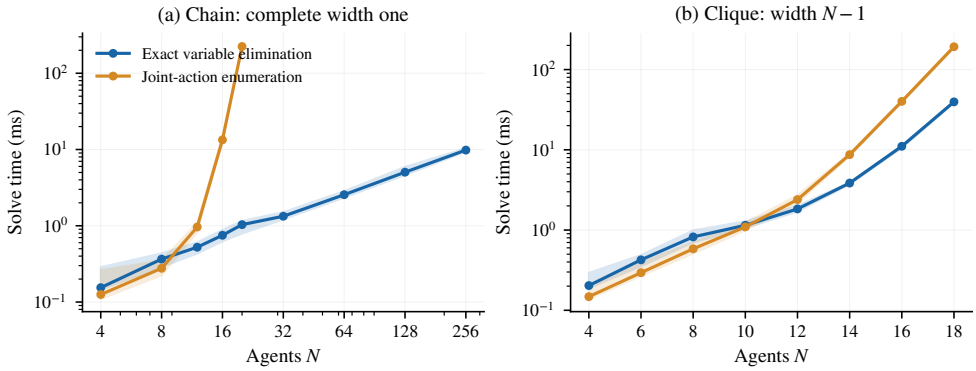


Figure 5: Exact planning on identical certified objectives. Complete action width is one on chains and $N - 1$ on cliques. Shading gives interquartile ranges over 25 instances. Enumeration is omitted above $N = 20$. The clique measurements concern generic elimination, not a specialized solver.

All 325 paired instances return identical scores, actions, and fallback decisions, with no infeasible outputs. At $N = 20$, median chain runtimes are 1.035 ms for elimination and 224.322 ms for enumeration. Elimination takes 9.824 ms at $N = 256$. The largest chain elimination table has four entries, whereas the $N = 18$ clique requires 262,144 entries and takes 39.575 ms. These counts are table entries, not measured memory. This clique admits a specialized $O(N)$ solver because at most one action can be active. The benchmark therefore measures generic elimination costs under different induced widths, not intrinsic hardness or a statistical advantage over centralized learning.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction match the assumptions and guarantees in Sections 3–5. Appendix K states the scope of the horizon-one learning study and planning benchmarks.

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